INDIAN ARMED FORCES (SD/SW)

CHAPTER AF V: ROLE & TASKS OF COMBAT ARMS, COMBAT SUPPORT ARMS AND SERVICES



TEACHING INSTRUCTIONS

Period	:	**02 (80 Mins)**
Type	:	**Lecture and Practice**
Year	:	**2nd Year SD/SW**
Conducting Officer	:	**Permanent Instructor**

Training Aids: Class room, OHP, Board and Chalk/ Markers

Time Plan

➢ **Introduction**	:	**05 Mins**
➢ **Part I**	:	**15 Mins**
➢ **Part II**	:	**15 Mins**
➢ **Part III**	:	**20 Mins**
➢ **Part IV**	:	**20 Mins**
➢ **Conclusion**	:	**05 Mins**

INTRODUCTION

1. The mission of the Indian Armed Forces is to ensure national security and unity, to defend the nation from external aggression and internal threats and to maintain peace and security within its borders. The primary role of Indian Army is to be the lead agency in fulfilling the mission as stated above against and land based external and internal threats ensuring sovereignty and territorial integrity of our country. In addition, when requisitioned, it conducts humanitarian rescue operations, relief operations during natural calamities and is employed in aid to civil authorities. India is also one of the largest troop contributors to the UN and to fulfil this mandate Indian Army takes part as Indian contingent in nominated UN Peace Keeping Missions.

2. The Indian Armed Forces are structured into different branches, each with specific responsibilities. Combat Arms units such as Infantry, Armoured Corps and Mechanised Infantry, are directly involved in engaging and defeating enemy forces. Combat Support arms provide essential support to combat units, including intelligence, communications, engineering and logistics. Services ensure the overall functioning and sustainability of military operations by providing medical care, supply chain management and other critical support functions.

3. Additionally, this chapter will cover some significant military operations conducted since independence, providing a historical context to the role performed by the Indian Armed Forces. The command-and-control structure, which ensures effective leadership and coordination within the military, will also be discussed. Finally, an overview of field formations, including Corps, Divisions and Brigades, will be provided to bring out the organizational hierarchy and deployment mechanics of the Indian Army.

PREVIEW

The lecture will be conducted in the following parts:

- **Part I: Role and Tasks of Combat Arms**.
- **Part II: Role and Task of Combat Support Arms**.
- **Part III: Role and Task of Services**
- **Part IV: Major Operations and peace-keeping missions since Independence**.
- **Part V: Command and Control of Field Formations**.

LEARNING OBJECTIVES

- **Role and task of all components of Indian army – Combat Arms, Combat Support Arms and Services**.
- **Details about major operations and missions of Indian Armed Forces**.
- **Their field formations and command & control setup**.

PART I: ROLE AND TASK OF COMBAT ARMS

4. Combat arms are the frontline units of the Indian Armed Forces, directly engaging in combat operations to achieve strategic and tactical objectives. These units are essential for both offensive and defensive operations, providing the necessary firepower, mobility and versatility to confront and defeat enemy forces. In the Indian Army, the primary branches of combat arms include the Armoured Corps, Infantry and Mechanised Infantry, each with distinct role and tasks that contribute to the overall accomplishment of any military objective.



5. **Armoured Corps**.

(a) **Role(Mobile and Offensive Action)**. The Armoured Corps is an organisation with tanks as the main fighting weapon platforms. It is designed to deliver powerful, rapid strikes against enemy positions, utilizing speed, firepower and protection to conduct swift and decisive attacks. Their role is crucial in manoeuvre warfare, where mobility and the ability to exploit enemy weaknesses by outflanking the enemy are key to success.

(b) **Tasks**.

(i) **Deep Penetration Operations**. Conducting operations that penetrate deep into enemy territory to disrupt command and control structures and cause destruction.

(ii) **Engaging Enemy Armoured Units**. Out manoeuvring and destroying enemy tanks and armoured vehicles.

(iii) **Providing Fire Support**. Offering direct fire support to Infantry and other units during offensive and defensive operations.

(iv) **Securing Key Terrain**. Capturing and holding strategic positions that are vital for the success of broader military objectives.

6. **Infantry**.

(a) **Role (Close Combat and Securing Ground)**. The Infantry is responsible for engaging enemy forces directly and maintaining control over strategic positions. They operate by fire and mov tactics, close in with the enemy and neutralise all opposition, capturing important ground enemy. Similarly, they also hold key location, repel any enemy and destroy the enemy by counter attack after weakening him. Since they operate on foot, they are versatile and capable of operating in any terrains and in any combat scenario.

(b) **Tasks**.

(i) **Patrols and Reconnaissance**. Conducting patrols to gather intelligence and maintain security in operational areas.

(ii) **Direct Combat Engagement**. Engaging enemy forces in close combat, utilizing a range of weapons and tactics.

(iii) **Capturing Positions**. Capturing important locations by attacking the enemy, close in by fire and move, neutralising enemy by accurate fire of direct and indirect firing weapons and capturing the objectives.

(iv) **Securing Positions**. Establishing and defending strategic positions and infrastructure.

(v) **Support Operations**. Providing support to other units through coordinated fire and manoeuvre tactics, ensuring the success of combined arms operations.

7. **Mechanised Infantry**.

(a) **Role (Enhanced Mobile Warfare and Securing Ground)**. The Mechanised Infantry enhances the mobility and protection of foot soldiers by equipping them with armoured vehicles such as Infantry Combat Vehicles (ICVs) or Armoured Personnel Carriers (APCs). These troops engage in combat utilising tactics from both armoured and Infantry units. This approach enables swift deployment, comparable to tanks and improves survival rates in battle zones. Mechanised Infantry serves as a link between conventional Infantry and armoured divisions, thereby increasing operational efficacy. Like Infantry, they can also close in with the objective, capture objective and hold it for limited time.

(b) **Tasks**.

(i) **Rapid Assaults**. Conducting swift assaults to capture and secure objectives, often in coordination with armoured units.

(ii) **Mobile Warfare**. Enhanced mobility and protection allow them to operate effectively in high-threat environments with armoured units. They manoeuvre with tanks, occupy advantageous locations to destroy enemy mechanise forces.

(iii) **Combined Arms Operations**. Supporting integrated operations with other combat arms, ensuring a cohesive and effective combat strategy.

(iv) **Mobile Fire Support**. Providing fire support to other combat arms enhancing their combat effectiveness.

This section highlights the distinct roles and tasks of the combat arms, emphasizing their contributions to the overall effectiveness and success of military operations. Each branch, with its unique capabilities, plays a vital role in achieving strategic objectives and maintaining national security.

DID YOU KNOW?

- Tanks are highly effective and destructive war fighting machines which is the main stay of any operation of Tank regiments. Modern tanks with their armour, mobility, protection and fire power are most suited for prevailing battle field conditions in open country and are widely used by all armies.
- Indian Army's Mechanised Infantry Regiment was raised on 02 April 1979, it is the newest regiment in the Indian Army and combines cutting-edge technology with military tradition. The Regiment has 27 Mechanised Infantry battalions and along with 21 battalions of the Brigade of the Guards makes up the Mechanised Infantry Forces of the country.
- Infantry Battalion is the basic unit of Infantry which has fighting companies, support companies and logistic companies which can fight and sustain itself in a battle field. A standard Indian Infantry Battalion has four Rifle Companies, one Support Company and One Head Quarter Company (Logistic Company). A group of similar Infantry units of Indian Army makes an Infantry Regiment like the Sikh Light Infantry Regiment. Indian Army has 25 Infantry Regiments



INSIGNIA OF SOME INDIAN INFANTRY REGIMENTS

ALL INFANTRY REGIMENTS OF INDIAN ARMY

Ser No	Regiment	Year of Raising
1.	PARA	1761
2.	PUNJAB	1705
3.	MADRAS	1758
4.	GRENADIERS	1796
5.	MARATHA LI	1768
6.	RAJ RIF	1817
7.	RAJPUT	1798
8.	JAT	1817
9.	SIKH	1846
10.	SIKH LI	1941
11.	DOGRA	1858
12.	GARH RIF	1901
13.	KUMAON (including NAGA)	1730

Ser No	Regiment	Year of Raising
14.	ASSAM	1941
15.	BIHAR	1941
16.	MAHAR	1941
17.	JAK RIF (including LADAKH SCOUTS)	1837
18.	JAK LI	1948
19.	1 GR	1815
20.	4 GR	1857
21.	3 GR	1815
22.	9 GR	1817
23.	5 GR	1858
24.	8 GR	1836
25.	11 GR	1948

PART II: ROLE AND TASK OF COMBAT SUPPORT ARMS

8. Combat support arms provide essential operational assistance to combat units, enhancing their effectiveness and ensuring the success of military operations. These units offer specialized support in areas such as fire support, air defence, reconnaissance, engineering, communications and intelligence. The primary branches of combat support arms include Artillery, Army Air Defence, Army Aviation Corps, Corps of Engineers, Corps of Signals and Intelligence Corps, each having distinct role and tasks.



Artillery Gun



Mobile Base Transceiver Station



Helicopter of AAC



Akash Surface to Air Missile

9. **Artillery**

(a) **Role**.

(i) **Fire Support**. Providing indirect fire to support combat operations.

(ii) **Area Denial and Force Protection**. Using artillery to deny enemy access to key areas and protect friendly forces.

(b) **Tasks**.

(i) **Indirect Fire Missions**. Delivering accurate and sustained firepower to neutralize or destroy enemy targets.

(ii) **Counter-Battery Fire**.Targeting and neutralizing enemy artillery units.

(iii) **Suppression of Enemy Air Defences (SEAD)**. Disabling enemy air defence systems to protect friendly aircraft.

(iv) **Smoke and Illumination**. Using smoke to obscure enemy vision and illumination rounds to enhance visibility for friendly forces.

DID YOU KNOW?

- The rate of fire for air defence (AD) is measured in rounds per minute (RPM). Here are some examples of some guns:-
- **M61 Vulcan**. This 20mm rotary cannon has a rate of fire of about 6,000 RPM.
- **Oerlikon GDF**. This 35mm twin-barrel autocannon has a rate of fire of about 1,100 RPM per barrel.
- **ZSU-23-4 Shilka**. This 23mm quad-barrel self-propelled anti-aircraft gun has a rate of fire of about 3,400 RPM.
- The Sarvatra Multi-Span Mobile Bridge System, developed by the Defence Research and Development Organisation (DRDO), is designed to bridge gaps of up to 75 meters using five scissor bridges, each with a span of 15 meters. This system allows combat battalions to cover significant distances by quickly deploying bridges over obstacles like rivers and ravines, ensuring mobility and operational efficiency in various terrains.

10. **Army Air Defence**

(a) **Role**. Air Defence to Mobile Forces and Vital Areas/Vital Points (VAs/VPs). Protecting mobile forces and critical infrastructure from aerial threats.

(b) **Tasks**.

(i) **Detection and Tracking**. Identifying and monitoring enemy aircraft and missiles using specialized radars.

(ii) **Engagement**. Using surface-to-air missiles and anti-aircraft guns to destroy or neutralize aerial threats.

(iii) **Coordination with Air Force**. Working closely with Air Force units to ensure comprehensive air defence coverage.

(iv) **Protection of Key Assets**. Defending vital areas and points such as command centres, logistics hubs and critical infrastructure.

11. **Army Aviation**

(a) **Role**. Reconnaissance and Observation. Conducting aerial reconnaissance and observation missions to gather intelligence and support ground operations.

(b) **Tasks**.

(i) **Air Assault**. Used for air assault on enemy bargets though attack helicopters.

(ii)) **Combat Support**. Provides combat support to fighting aircraft.

(iii) **Aerial Surveillance**. Monitoring enemy movements and positions from the air.

(iv) **Target Acquisition**. Identifying and designating targets for artillery and other combat units.

(v) **Medical Evacuation (MEDEVAC)**. Providing rapid evacuation of wounded personnel from the battlefield.

(vi) **Logistical Support**. Transporting troops, equipment and supplies to and from operational areas.

12. **Corps of Engineers**

(a) **Role**. Fortifications, Demolitions, Track Work and Obstacle Creation/Removal. Enhancing battlefield mobility and survivability through engineering tasks.

(b) **Tasks**.

(i) **Construction of Defensive Positions**. Building fortifications, bunkers and other defensive structures.

(ii) **Demolitions**. Conducting controlled demolitions to destroy enemy obstacles and infrastructure.

(iii) **Obstacle Creation and Removal**. Placing and clearing mines, barriers and other obstacles to control movement on the battlefield.

(iv) **Bridging Operations**. Constructing temporary bridges to enable the movement of troops and vehicles across water obstacles.

13. **Corps of Signals**

(a) **Role**. Communication in Orders and Instructions and ensuring reliable and secure communication for command and control of military operations.

(b) **Tasks**.

(i) **Establishing Communication Networks**. Setting up and maintaining communication systems for voice, data and video transmission.

(ii) **Signal Intelligence (SIGINT)**. Intercepting and analysing enemy communications.

(iii) **Electronic Warfare (EW)**. Disrupting enemy communications and protecting friendly communication networks.

(iv) **Cyber Operations**. Defending against cyber threats and conducting offensive cyber operations.

14. **Intelligence Corps**

(a) **Role**. Gathering, Analysing and Disseminating Military Information. Providing actionable intelligence to support decision-making and operations.

(b) **Tasks**.

(i) **Human Intelligence (HUMINT)**. Collecting information from human sources.

(ii) **Imagery Intelligence (IMINT)**. Analysing aerial and satellite imagery.

(iii) **Signals Intelligence (SIGINT)**. Intercepting and analysing enemy communications and electronic signals.

(iv) **All-Source Analysis**. Integrating information from various sources to produce comprehensive intelligence reports.

(v) **Dissemination**. Distributing intelligence to commanders and units in a timely manner.

15. This section highlights the distinct roles and tasks of the combat support arms, emphasizing their contributions to the overall effectiveness and success of military operations. Each branch, with its specialized capabilities, plays a vital role in supporting combat units and achieving strategic objectives.

PART III: ROLE AND TASK OF SERVICES

16. Services are essential for the smooth functioning and sustainability of military operations. These units provide critical logistical, medical, engineering and administrative support, ensuring that combat and combat support arms can perform their duties effectively both during peace

and war. The primary branches of supporting services with their role and tasks are given as under:-









17. **Electrical and Mechanical Engineering (EME)**

(a) **Role**. Maintenance and Repair and ensuring the operational readiness of military equipment through maintenance and repair.

(b) **Tasks**.

(i) **Design and Development**. Creating and testing electrical and mechanical systems and components.

(ii) **Maintenance**. Performing routine and emergency maintenance on vehicles, weapons and other equipment.

(iii) **Troubleshooting**. Diagnosing and fixing mechanical and electrical issues.

(iv) **Upgrades and Modifications**. Implementing upgrades and modifications to enhance equipment performance.

18. **Army Service Corps (ASC)**

(a) **Role**. Logistical Support and providing essential supplies and transportation to sustain military operations.

(b) **Tasks**.

(i) **Supply Chain Management**. Managing the procurement, storage and distribution of food, fuel and other supplies.

(ii) **Transportation**. Coordinating the movement of troops, equipment and supplies using various modes of transport.

(iii) **Field Services**. Offering services such as laundry, bakery and water supply in operational areas.

19. **Army Ordnance Corps**

(a) **Role**. Supplyand maintenance of Ordnance and managing the supply and maintenance of weapons, ammunition and military equipment.

(b) **Tasks**.

(i) **Storage and Distribution**. Ensuring the safe storage and timely distribution of ordnance stores.

(ii) **Maintenance and Repair**. Conducting maintenance and repair of weapons and equipment.

(iii) **Inventory Management**. Keeping accurate records of ordnance inventory and ensuring accountability.

20. **Army Medical Corps (AMC)**

(a) **Role**. Provide Medical Support and comprehensive medical care to military personnel.

(b) **Tasks**.

(i) **Emergency Care**. Delivering immediate medical care on the battlefield.

(ii) **Routine Treatment**. Offering routine medical services at health centres and hospitals.

(iii) **Health Promotion**. Conducting health promotion and disease prevention programs.

(iv) **Medical Evacuation**. Coordinating the evacuation of injured personnel from combat zones.

21. **Army Dental Corps**

(a) **Role**. Dental Health and ensuring the dental health of military personnel.

(b) **Tasks**.

(i) **Dental Care**. Providing routine and emergency dental care.

(ii) **Preventive Dentistry**. Conducting dental health education and preventive care programs.

(iii) **Field Support**. Offering dental services in operational areas to maintain dental readiness.

22. **Military Nursing Service (MNS)**

(a) **Role**. Providing Nursing Care. Providing nursing care to military personnel and their families.

(b) **Tasks**.

(i) **Patient Care**. Delivering high-quality nursing care in Military Hospitals and field units.

(ii) **Health Education**. Educating military personnel on health and wellness.

(iii) **Support in Operations**. Assisting in medical operations and humanitarian missions.

23. **Remount and Veterinary Corps**

(a) **Role**. Animal Care and managing the health and welfare of military animals.

(b) **Tasks**.

(i) **Veterinary Services**. Providing medical care to military animals.

(ii) **Animal Training**. Training animals for various military roles.

(iii) **Breeding and Remounting**. Managing breeding programs and remounting operations.

24. **Army Education Corps (AEC)**

(a) **Role**. Provide Educational Support and enhancing the educational and professional development of military personnel.

(b) **Tasks**.

(i) **Training Programs**. Conducting educational and training programs.

(ii) **Career Development**. Providing career counselling and development services.

(iii) **Literacy and Language Training**. Offering literacy and language training to improve communication skills.

25. **Corps of Military Police**

(a) **Role**. Ensure Law Enforcement and Military Policing and maintaining law and order within the military.

(b) **Tasks**.

(i) **Security Operations**. Conducting security operations to protect military installations and personnel.

(ii) **Traffic Control**. Managing traffic and transportation within military areas.

(iii) **Investigation**. Investigating crimes and incidents involving military personnel.

26. **Army Postal Service**

(a) **Role**. Provide postal services to military personnel during peacetime and operations.

(b) **Tasks**.

(i) **Mail Processing and Distribution**. Process, transport and distribute personal and official mail.

(ii) **Special Services**. Handle special services like registered mail and money orders.

(iii) **Efficient Mail Operations**. Ensure efficient mail operations in various locations, including remote and conflict areas.

27. **Judge Advocate General Branch (JAG)**.

(a) **Role**. Legal Support. Providing legal services and advice to the Indian Armed Forces.

(b) **Tasks**.

(i) **Legal Advice**. Offering legal advice on military operations and personnel matters.

(ii) **Military Justice**. Administering military justice and conducting court-martials.

(iii) **Legal Education**. Educating military personnel on legal issues and regulations.

28. **Military Engineering Services (MES)**.

(a) **Role**. Infrastructure Development and constructing and maintaining military infrastructure.

(b) **Tasks**.

(i) **Construction Projects**. Building and maintaining military facilities and infrastructure.

(ii) **Utilities Management**. Managing utilities such as water, electricity and sewage.

(iii) **Environmental Management**. Ensuring environmental compliance and sustainability in military operations.

29. **Border Roads Organization (BRO)**.

(a) **Role**. Road Construction and Maintenance and developing and maintaining road networks in border areas.

(b) **Tasks**.

(i) **Road Construction**. Building roads to improve connectivity and mobility in border regions.

(ii) **Maintenance**. Maintaining and repairing existing road networks.

(iii) **Strategic Projects**. Undertaking strategic infrastructure projects to support military operations.

30 **Army Pioneer Corps**.

(a) **Role**. Barrier Support and providing labour support for various military tasks.
(b) **Tasks**.

(i) **Construction Support**. Assisting in construction and engineering projects.

(ii) **Logistical Support**. Providing manpower for logistical operations.

(iii) **Field Operations**. Supporting field operations with labour-intensive tasks.

31. This section highlights the distinct roles and tasks of the supporting services, emphasizing their contributions to the overall effectiveness and sustainability of military operations. Each branch, with its specialized capabilities, plays a vital role in supporting combat and combat support arms and achieving strategic objectives.

PART IV: MAJOR OPERATIONS AND PEACE KEEPING MISSIONS SINCE INDEPENDENCE

32. Since independence, the Indian Armed Forces have been involved in numerous significant operations, both within and outside the country. These operations have been crucial in defending national sovereignty, maintaining internal security and contributing to international peacekeeping efforts. This section provides an overview of some of the major operations and peacekeeping missions conducted by the Indian Armed Forces.

33. **Major Operations**.

(a) **Indo-Pakistan War of 1947-1948**.

(i) **Objective**. To defend Jammu and Kashmir from Pakistani tribal invaders and regular forces.

(ii) **Outcome**. It resulted in eviction of the Pakistani tribal and regular forces in valleys from large areas of the princely state of Jammu and Kashmir and

establishment of the Line of Control (LoC) and the accession of princely state to India.

(b) **Sino-Indian War (1962)**.

(i) **Objective**. To defend against Chinese incursions in the Aksai Chin and Arunachal Pradesh regions.

(ii) **Outcome**. The war ended with universe withdrawal to its pre-war position, creation of Line of Actual Control and also highlighted the need for modernization and restructuring of the Indian Armed Forces.



Rifle-toting Indian soldiers on patrol during the brief, bloody 1962 Sino-Indian border War

(c) **Indo-Pakistan War of 1965**.

(i) **Objective**. To counter Pakistani aggression in the Rann of Kutch and Jammu and Kashmir.

(ii) **Outcome**. Even though Pakistan never formally conceded defeat, the loss of strategic Haji Pir Pass and threatening of the major Pakistani city of Lahore forced them to agree to a United Nation brokered ceasefire.

(d) **Indo-Pakistan War of 1971**.

(i) **Objective**. To support the independence movement in East Pakistan (now Bangladesh) and counter Pakistani military actions.

(ii) **Outcome**. The creation of Bangladesh, capture of Prisoners of War (PoW) of Pakistani Army and a decisive victory for India, significantly established Indian supremacy in the geopolitical landscape of South Asia.

(e) **Kargil War (1999)**.

(i) **Objective**. To evict Pakistani Army intruders from the Kargil sector in Jammu and Kashmir.

(ii) **Outcome**. A decisive victory for India, with the successful recapture of key positions and the restoration of the LoC.



Soldiers of the Indian Army depicted after capturing a hill from Pakistani forces during the Kargil War

34. **United Nations Peace-Keeping Missions**. India has been a significant contributor to United Nations peacekeeping missions since 1948. Here are some of the key peacekeeping missions that India was and is currently part of:-

(a) **Completed Peacekeeping Missions**.

(i) **United Nations Mission in Sierra Leone (UNAMSIL)**. Indian peacekeepers were involved in this mission from 1999 to 2005 to help stabilize Sierra Leone.

(ii) **United Nations Mission in Ethiopia and Eritrea (UNMEE)**. India contributed to this mission from 2000 to 2008 to monitor the ceasefire between Ethiopia and Eritrea.

(iii) **United Nations Stabilization Mission in Haiti (MINUSTAH)**. Indian peacekeepers have been part of this mission since December 1997, helping to stabilize the country.

(iv) **United Nations Mission in Liberia (UNMIL)**. Since April 2007, Indian troops have been contributing to the peacekeeping efforts in Liberia.

(b) **Ongoing Peacekeeping Missions**. Some Important UN Msn having significant Indian participation are as under:-

(i) **United Nations Interim Force in Lebanon (UNIFIL)**. Since December 1998, Indian troops have been part of this mission to maintain peace and security in Lebanon.

(ii) **United Nations Organization Stabilization Mission in the Democratic Republic of the Congo (MONUSCO)**. India has been involved since January 2005, contributing to the stabilization efforts in the region.

(iii) **United Nations Mission in South Sudan (UNMISS)**. Indian peacekeepers have been part of this mission since April 2005, helping to protect civilians and support the peace process.

(c) **Relief Operations**.

(i) **Wayanad Landslides, Kerala (2024)**. The Indian Army, National Disaster Response Force (NDRF) and other agencies conducted large-scale rescue operations, saving over 4,000 people and relocating more than 10,000 to relief camps.

(ii) **Cyclone Amphan (2020)**. A super cyclone that caused extensive damage in West Bengal and Odisha. The Indian Armed Forces and NDRF were involved in rescue and relief operations, providing essential supplies and medical aid.

(iii) **Operation Surya Hope (2013)**. Launched by Indian Army to provide relief and rescue operations in the Badrinath, Hemkund and Kedarnath regions. This operation was crucial in evacuating thousands of stranded pilgrims and providing essential supplies to those affected by the floods in Uttarakhand.

(iv) **Operation Ajay (2023)**. Launched to facilitate the return of Indians from Israel during the conflict with Hamas militants

(v) **Operation Dost (2023)**. A search and rescue operation to aid Syria and Turkey after the devastating earthquake in February 2023.

35. India's commitment to UN peacekeeping missions and relief operations underscores its dedication to global peace and security, with Indian peacekeepers being recognized for their professionalism, dedication and bravery in some of the most challenging environments around the World.

36. Command and control (C2) are a critical aspect of military operations, ensuring effective leadership, coordination and execution of missions. It involves the exercise of authority and direction by commanders over assigned forces in the accomplishment of their mission. This section will explore the structure and mechanisms of command and control within the Indian Armed Forces.

37. **Command Structure**.

(a) **National Level**.

(i) **President of India**. The Supreme Commander of the Indian Armed Forces.

(ii) **Ministry of Defence (MoD)**. Responsible for the overall defence policy and administration.

(iii) **Chiefs of Staff Committee (COSC)**. Comprising the Chief of Defence Staff (CDS) and the Chiefs of the Army, Navy and Air Force, this committee provides strategic guidance and coordination.

(b) **Strategic Level**.

(i) **Integrated Defence Staff (IDS)**. Assists the COSC in joint planning and operations.

(ii) **Theatre Commands**. Proposed joint commands that integrate the capabilities of the Army, Navy and Air Force in specific geographical areas.

(iii) **Army Commands**. Seven commands, each headed by a General Officer Commanding-in-Chief (GOC-in-C), responsible for operational readiness and execution.

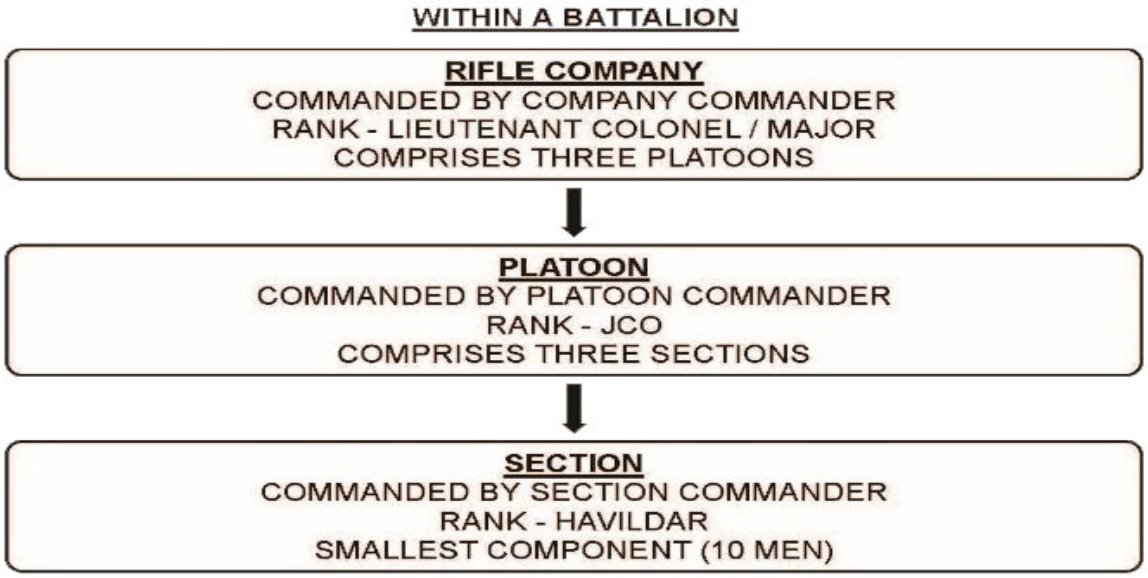
(iv) **Navy Commands**. Two operational commands, both headed by the Flag Officer Commanding-in-Chief (FOC-in-C), responsible for operating missions above, under or on seas.

(v) **Air Force Commands**. Five operational commands both headed by the Air Officer Commanding-in-Chief (AOC-in-C), responsible air missions.

(c) **Operation Command/Tactical Command**. Corps, Divisions and Brigades: Subordinate formations that execute tactical operations under the direction of their respective commands.

38. **Field Formations**. Field formations are the organizational structures that enable the Indian Armed Forces to conduct operations effectively. These formations range from large corps to smaller brigades, each with specific roles and responsibilities. For the ease of understanding, the same is evaluated by a flow diagram.





CONCLUSION

39. The Indian Armed Forces are integral to the nation's security, performing a wide range of roles from direct combat to logistical and medical support. Combat arms, including the Armoured Corps, Infantry and Mechanised Infantry, are at the forefront of military operations, while combat support arms and supporting services provide essential assistance and sustainment.

40. Through major operations since independence, the Indian Armed Forces have demonstrated their capability and resilience. The command-and-control structure ensures effective leadership and coordination, while field formations enable efficient operational execution.

41. In essence, the diverse branches and specialized units of the Indian Armed Forces work in unison to protect the nation and maintain peace and stability, adapting to evolving challenges and ensuring readiness for future operations. Their continued evolution and commitment to excellence underscore their vital role in safeguarding the nation's interests and contributing to global peacekeeping efforts.

SUMMARY

The Indian Armed Forces ensure national security and support civil authorities.

Combat Arms.

- **Armoured Corps**. Mobile and offensive actions; tasks include deep penetration and fire support. They play a crucial role in manoeuvre warfare.
- **Infantry**. Close combat and securing ground; tasks include patrols and direct engagement. They are versatile and operate in diverse terrains.

- **Mechanised Infantry**. Enhanced mobility and protection; tasks include rapid assaults and mobile fire support. They integrate with armoured units for coordinated operations.

Combat Support Arms.

- **Artillery**. Provides fire support and area denial to protect and assist combat units.
- **Army Air Defence**. Protects mobile forces and vital areas from aerial threats.
- **Army Aviation Corps**. Conducts reconnaissance and observation to gather intelligence.
- **Corps of Engineers**. Handles fortifications, demolitions and obstacle management.
- **Corps of Signals**. Ensures communication for command and control.
- **Intelligence Corps**. Gathers, analyses and disseminates military information

Supporting Services.

- **Maintenance and Logistics**. Electrical and Mechanical Engineering, Army Service Corps, Army Ordnance Corps ensure operational readiness and supply management.
- **Medical Support**. Army Medical Corps, Army Dental Corps, Military and Nursing Service provide comprehensive healthcare.
- **Specialized Support**. Remount and Veterinary Corps, Army Education Corps, Corps of Military Police, Judge Advocate General Branch, Military Engineering Services, Border Roads Organization, Army Pioneer Corps offer various specialized services.

MAJOR OPERATIONS

- Key operations include the Indo-Pakistan wars, Sino-Indian War and Kargil War. These operations have demonstrated the strategic importance and resilience of the Indian Armed Forces, showcasing their ability to adapt and respond to diverse challenges. Each operation has contributed to the evolving tactics and strategies of the military.
- India has also been a significant contributor to United Nations peacekeeping missions and Relief operations since its independence. Same important peace-keeping and relief missions United Nations Mission in Sierra Leone (UNAMSIL), United Nations Mission in Ethiopia and Eritrea (UNMEE), United Nations Interim Force in Lebanon (UNIFIL), United Nations Organization Stabilization Mission in the Democratic Republic of the Congo (MONUSCO), United Nations Mission in South Sudan (UNMISS), United Nations Stabilization Mission in Haiti (MINUSTAH), United Nations Mission in Liberia (UNMIL)

- **Field Formations**.

  - Corps. Operational command coordinating multiple divisions.
  - Divisions. Tactical command supporting brigades.
  - Brigades. Direct combat operations and support to battalions.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What is the primary role of the Armoured Corps?

(a) Close combat
(b) Mobile and offensive action
(c) Reconnaissance
(d) Medical support

Q2. Which branch of the Indian Armed Forces is responsible for providing fire support and area denial?

(a) Infantry
(b) Artillery
(c) Army Aviation Corps
(d) Corps of Engineers

Q3. What is the main task of the Army Air Defence?

(a) Fortifications and demolitions
(b) Air defence to mobile forces and VAs/VPs
(c) Gathering and analysing military information
(d) Logistical support

Q4. Which unit is responsible for reconnaissance and observation?

(a) Armoured Corps (b) Infantry
(c) Army Aviation Corps (d) Corps of Signals

Q5. What is the role of the Corps of Engineers?

(a) Communication in command and control
(b) Fortifications, demolitions and obstacle creation/removal
(c) Medical support
(d) Legal support

Q6. Which branch handles the maintenance and repair of military equipment?

(a) Army Service Corps
(b) Army Ordnance Corps
(c) Electrical and Mechanical Engineering
(d) Corps of Military Police

Q7. What is the primary task of the Army Medical Corps?

(a) Dental care (b) Veterinary services
(c) Medical support (d) Educational support

Q8. Which branch is responsible for constructing and maintaining

(a) military infrastructure?
(b) Military Engineering Services
(c) Corps of Engineers
(d) Border Roads Organization
(e) Army Pioneer Corps

Q9. Which operation led to the creation of Bangladesh?

(a) Operation Polo
(b) Operation Blue Star
(c) Indo-Pakistan War of 1971
(d) Kargil War

Q10. What is the role of the Corps of Signals?

(a) Providing fire support
(b) Ensuring communication in command and control
(c) Conducting reconnaissance
(d) Managing logistics

Q11. Which branch is responsible for law enforcement and security within the military?

(a) Corps of Military Police
(b) Judge Advocate General Branch
(c) Army Education Corps
(d) Border Roads Organization

Q12. Which operation was conducted to aid Syria and Turkey?
(a) Operation Ajay
(b) Operation Dost
(c) Operation Pawan
(d) Operation Parakram

Q13. What is the primary role of the Intelligence Corps?

(a) Providing medical care
(b) Gathering, analysing and disseminating military information
(c) Conducting demolitions
(d) Managing road construction

Q14. Which branch provides logistical support, including supply chain management and transportation?

(a) Army Ordnance Corps
(b) Army Service Corps
(c) Electrical and Mechanical Engineering
(d) Corps of Engineers

MILITARY COMMUNICATION

3

CHAPTER WISE INDEX: MILITARY COMMUNICATION

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MILITARY COMMUNICATION (SD/SW)

CHAPTER MC I: TYPES OF COMMUNICATION

"The telephone gives us the happiness of being together yet safely apart (Mason Cooley)"



TEACHING INSTRUCTIONS

Period	**:**	**01 (40 Mins)**
Type	**:**	**Lecture**
Year	**:**	**1st Year SD/SW**
Conducting Officer	**:**	**Permanent Instructor**

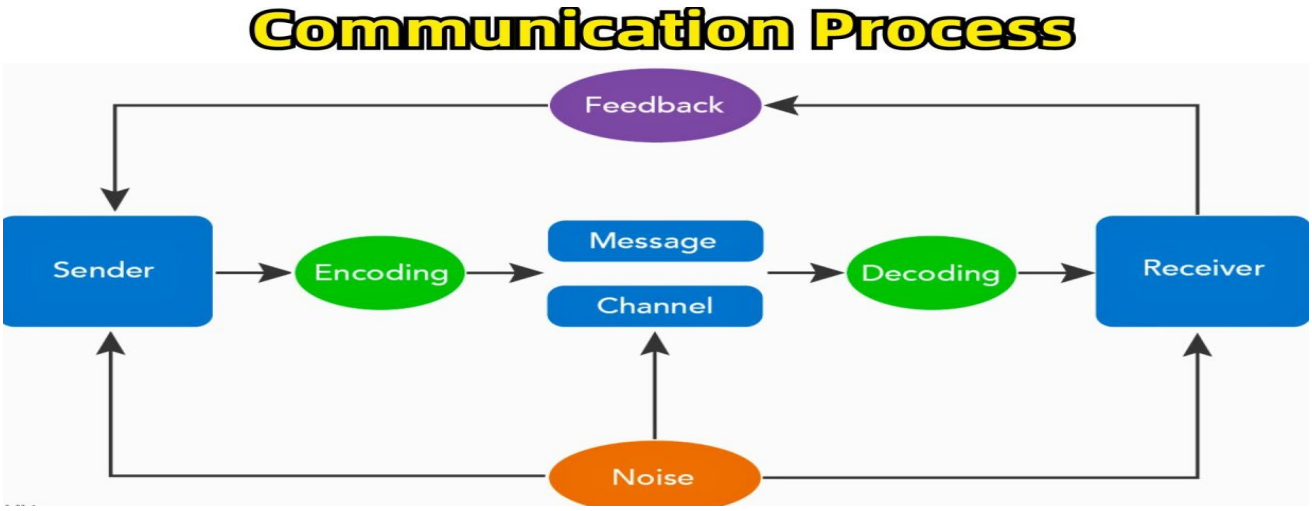
Training Aids: Class Room, Computer with OHP, Screen, Pointer Staff, Presentation, Script/Book/Lesson Plan, Board, and Markers

Time Plan

- **Introduction : 02 Mins**
- **Part I : 08 Mins**
- **Part II : 18 Mins**
- **Part II : 10 Mins**
- **Practice : 30 Mins**
- **Conclusion and Summary : 02 Mins**

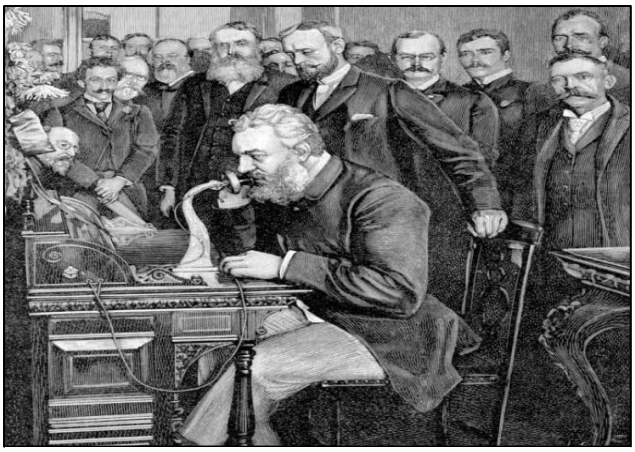
INTRODUCTION

1. Communication is the process where a sender shares a message with a receiver. First, the sender thinks of a message and decides how to express it (this is called encoding). Then, the message travels through a medium, like speaking, writing, or using technology. The receiver gets the message, interprets it (decoding), and gives feedback, like asking a question or responding. However, there are barriers that can make communication difficult, such as noise, language differences, emotions, or misunderstandings, which can interfere at any point in the process and affect how well the message is understood.



Flow Chart of Communication

2. Up till 1870's the people as well as the militaries of the world used dispatch riders, pigeons, light signals like light house and sound signals like drums to communicate over large distances. The invention of the telephone in 1876 by Graham Bell revolutionized the world of communications as individuals could speak directly to each other over large distances instantaneously. This brought in the era of line communication and morse code in the militaries. Subsequently, with the introduction of wave-based communications Radio sets were introduced in the Armies which remained the main stay of military communication until recently wherein digitization, internet and satellite-based communication have revolutionized the way people communicate both in the civil as well as in military domains.



First Communication on Telephone



Army Solider communicating on Radio set

3. The Military Communications refers to all aspects of communication or transaction of information by the Armed Forces including line, radio and digital media in battle and non-battle conditions. It includes the infrastructure, equipment, and protocols used by the military to facilitate communication between different sub units, units and command levels during operations or during peace. The primary purpose of any good military communication system is to ensure secure, reliable, and efficient communication across various domains, enabling coordination, command, and control. In this chapter the cadets will learn about the basics of military communications including line and radio communication.

INTERESTING FACTS

- **Indian Army Corps of Signals is the primary agency of Indian Army, which handles its military communications. It was formed on 15 February 1911 as a separate entity under Lieutenant Colonel S H Powell.**



PREVIEW

The lecture will be conducted in the following parts:

- **Part I: Line Communication.**
- **Part II: Radio Communication.**
- **Part III: Types of Radio Communication.**

LEARNING OBJECTIVES

- **Line communication - Basic components and advantages and disadvantages.**
- **Radio communication - Constituents, types of waves and propagation of waves.**
- **Advantages & disadvantages of net radio communication**
- **Advantages & disadvantages of radio relay communication.**

4. Some communication systems and equipment designed for use by defence forces are as under:

(a) High-frequency and very high frequency fixed and mobile radio communications.

(b) Ultra-high frequency combat net radios and accessories.

(c) Satellite based communication and navigation systems.

(d) Military neck microphones, handsets, headsets, and speakers.

(e) Military broadband or military intranet-based communication systems.

(f) Various types and format of secure line based, point to point communication systems.

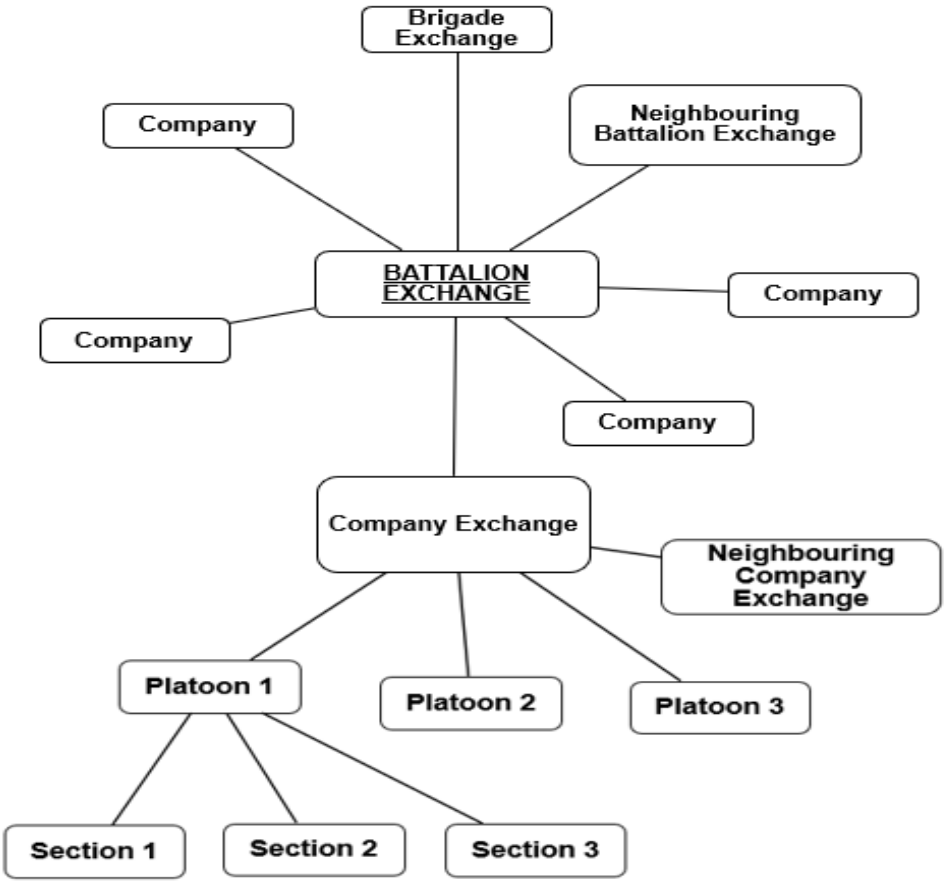
PART I: LINE COMMUNICATION

5. Line-based communication refers to the transfer of information between two or more people using a physical medium, like wires or cables. This type of communication relies on a direct, fixed connection, such as telephone lines or internet cables, to send messages, data, or signals. It is often used in systems like landline phones or broadband internet, where the message travels along a specific path between the sender and receiver. A telephone is best and most secure means of signal communication between individuals. A modern-day line communication set up requires exchanges, trunk media connecting exchanges, and local loop/cable to extend telephone connections to subscribers. A standard line communication set up in battle condition in any army unit will have about 30-40 telephone sets which are connected with an exchange and the range of communication is about 8 to 10 Kms. The telephone cables are laid on the ground connecting all subscribers to an exchange. Similarly, two exchanges can also be connected with line to enhance the subscriber base and range. Primarily, this is the basic means of signal communications for a static forces or when the forces are not moving.

Components of Line Communication

6. **Components of Line Communication**. These are sender, receiver, communication medium, the message to be communicated, and certain rules called protocols to be followed during communication. The components of line communication are as under:-

LAYOUT OF A BASIC LINE COMMUNICATION NETWORK



7. **Components of a Magneto Telephone set**. There are many types of simple magneto telephones, which are used by Indian Army for setting up line communication. The ringer arm of the magneto telephone is rotated to generate a ring at the exchange. Thereafter communication is made through the exchange with the required subscriber. To communicate pressel switch of the handset is pressed before speaking. The main components of a simple magneto telephone are as under:-

(a) Leather case with carrying strap.

(b) Generator handle.

(c) L1 and L2 Terminals.

(d) Handset.

(e) Microphone and Receiver.

(f) Pressel switch.

(g) Cord.

(h) Terminals for hand set.

(j) Socket for Hand set.

(k) Cable switch for CB Exchange.

(l) Battery Compartment.



Advantages and Disadvantages of Line Communication

8. **Advantages**.

(a) Reliable and practically free from electrical interference.

(b) Relatively secure.

(c) The number of circuits and message-carrying capacity is more but limited only by the availability of material and manpower.

9. **Disadvantages**.

(a) Vulnerable to physical interference and enemy interception along the entire length of the route.

(b) Takes time to construct.

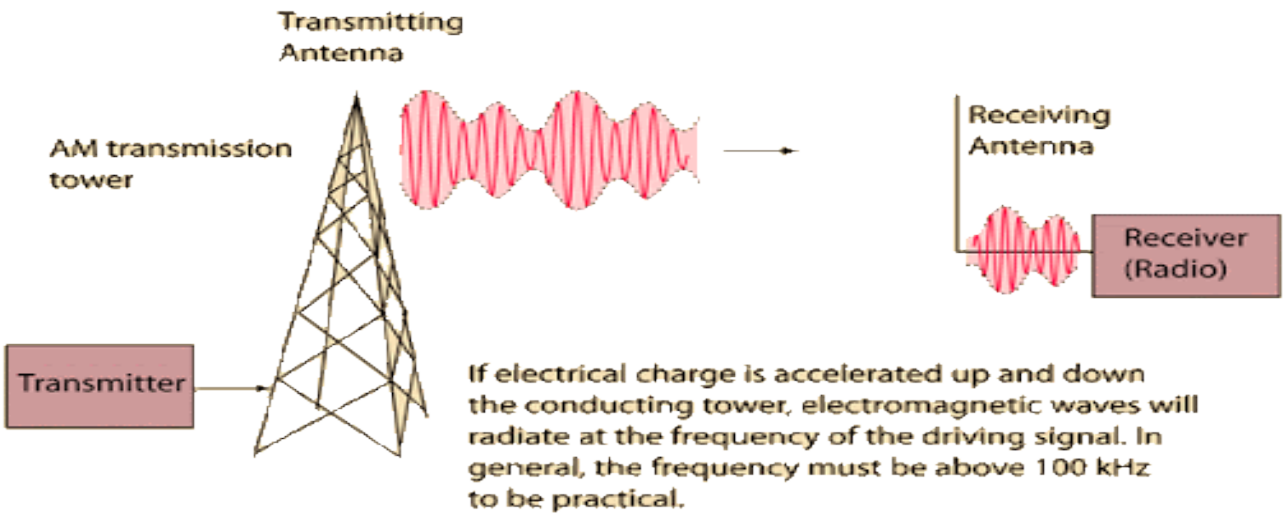
(c) Inflexible once it is laid.

(d) Expensive in men and material.

PART II: RADIO COMMUNICATION

10. **Components of Radio Communication**. Radio is the technology of using radio waves to carry information such as sound, by systematically modulating properties of electromagnetic energy waves. The information in the waves can be extracted when radio waves strike an electrical conductor and transform it back into its original form. Radio communication requires the use of both transmitting and receiving equipment. The transmitting equipment, which includes a radio transmitter and transmitting antenna and is

installed at the point from which messages are transmitted. The receiving equipment, which consists of a radio receiver and receiving antenna, is installed at the point at which messages are received.



11. The Indian Army utilizes various components in its radio communication systems to ensure efficient and secure communication for operations. The key components of radio communication in the Indian Army are as follows:-

(a) **Radio Transmitters and Receivers**. Transmitters are used to send signals across different frequencies. Receivers are used to receive signals from other communication devices. These devices can be portable (for infantry units), vehicle-mounted (for armored or mechanized units), or fixed (for headquarters or base stations).

(b) **Communication Radios**. The Indian Army uses different types of radios based on the operational requirement:

Type of Frequencies	Ranges	Remarks
HF (High Frequency) Radios.	Beyond 50-100 km	Long-range communication
UHF (Ultra High Frequency) Radios	Few Kilometers	Short-range communication
VHF (Very High Frequency) Radios.	between 30 to 50 km.	Communication between tactical units on the battlefield

(c) **Antennas**. Antennas play a crucial role in radio communication by transmitting and receiving signals. Depending on the type of radio, antennas vary in size and configuration:

(i) Whip Antennas: Common in handheld radios.

(ii) Long Wire Antennas: Used for HF radios.

(b) **Dish Antennas**. Used for SATCOM and high-frequency communications.

(c) **Radios in Vehicles**. Vehicles such as armored vehicles, trucks, and command centers are equipped with radios for communication during mobility

operations. These radios are often mounted in the vehicle and provide a stable communication link, even while moving.

Types of Wave

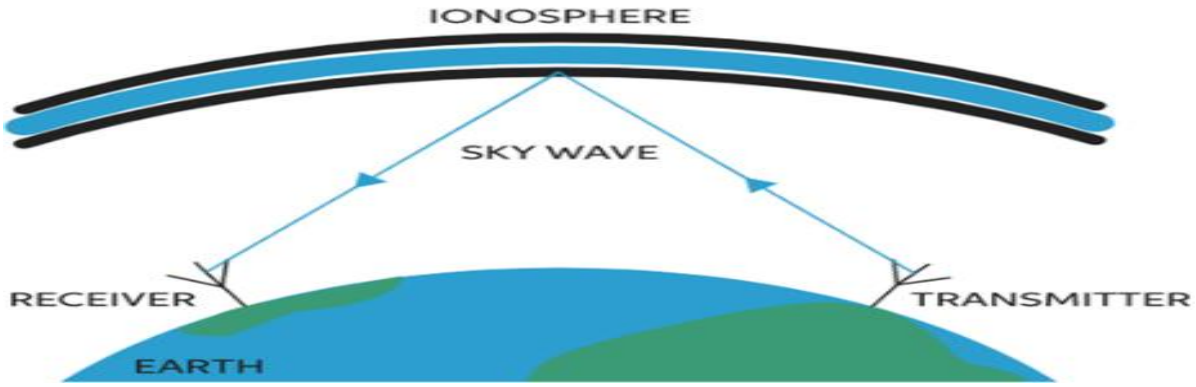
12. A wave can be described as a disturbance that travels through a medium from one location to another location. Types of waves can broadly be divided in two parts:-

(a) **Mechanical Waves**. A mechanical wave is a wave that is an oscillation of matter and therefore transfers energy through a medium.

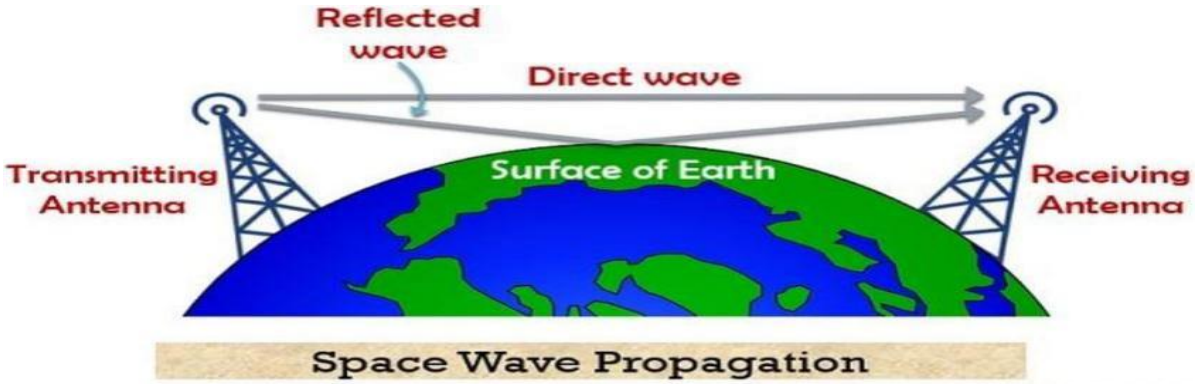
(b) **Electromagnetic Waves**. Electromagnetic waves are waves that can travel through the vacuum of outer space.

13. **Propagation of Wave**. The mode of propagation of electromagnetic waves (EMW) from transmitter to receiver depends upon the frequency employed. These can be of the following types:-

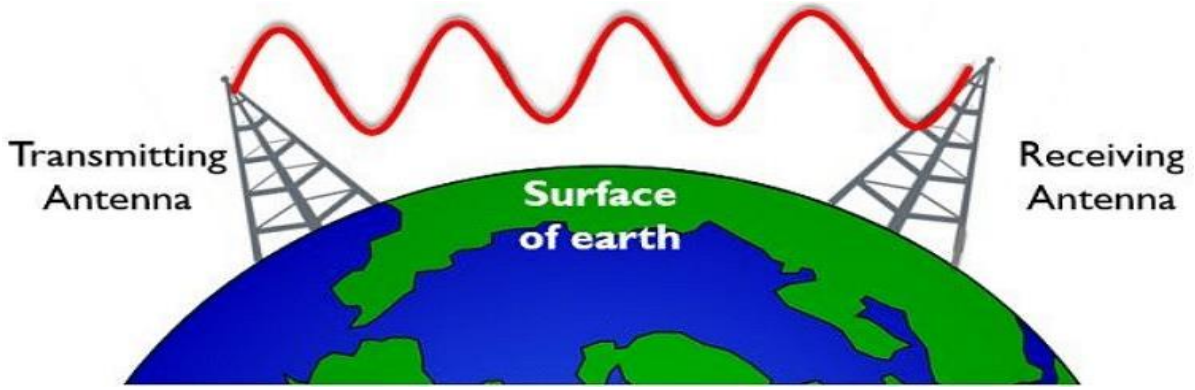
(a) **Sky Wave Propagation**. The sky waves are the radio waves that use an ionosphere layer existing to a height of 150 - 200 km from the surface of the earth and are reflected by the ionosphere of the earth's atmosphere.



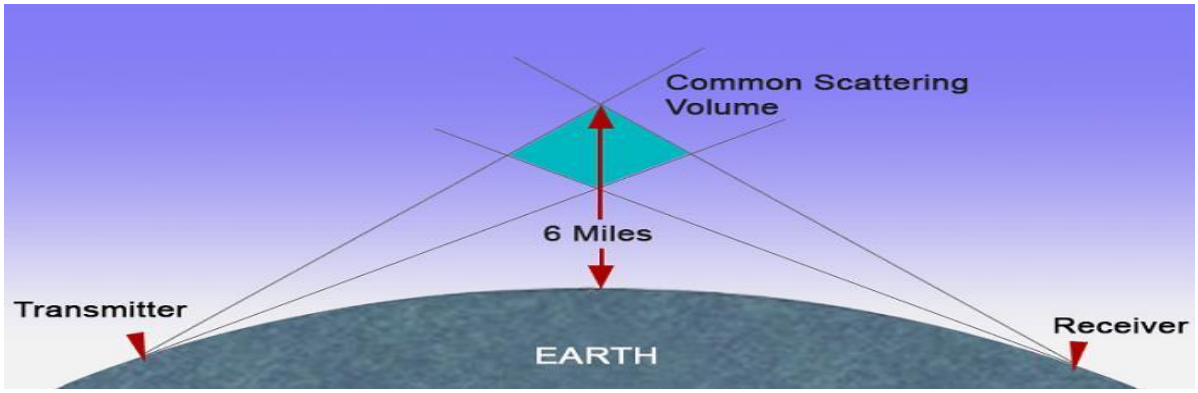
(b) **Space Wave Propagation**. Space waves can travel through the atmosphere from the transmitter antenna to the receiver antenna either directly or after reflection from the ground in the earth's troposphere region. Their range is limited by the curvature of the earth; therefore, the distance between two neighboring stations is approximately 50 Km.



(c) **Ground Wave Propagation**. It is a method of radio frequency propagation that uses the area between the surface of the earth and the ionosphere for transmission. Ground wave radio signal propagation is ideal for relatively short-distance propagation on these frequencies during the daytime with a limited range of approximately 30 km.



(d) **Tropospheric Scatter**. (Also known as troposcatter) It is a method of communicating with microwave radio signals over considerable distances – often up to 300 km and further depending on terrain and climate factors. Radio signals are transmitted in a narrow beam aimed just above the horizon in the direction of the receiver station. Communication distances are limited by the visual horizon to around 30–40 miles (48–64 km).



Radio Communication in Army

14. Radio communication in army is carried out in two ways namely Net Radio, which a number of subscribers are out station on a particular frequency and the communication is supervised by a Control. This type of communication is limited to a particular area and number of subscribers. However, the second way is Radio Relay, wherein a repeater radio set is used to enhanced the range of communication of two Net Radio networks.

15. **Net Radio**. Net radio is the basic signal communication for any mobile force. The efficiency of net radio communication is appreciably affected by factors such as weather, terrain, power output of the set, state of training of operators, and equipment maintenance. It provides facilities for the following:-

(a) **Radio Telephony**. Simple one-way voice communication, depending on the type of equipment available.

(b) **Data Transmission**. With the use of software and specially designed Laptops.

16. **Advantages of Net Radio**.

(a) Vulnerable only at the terminal and is therefore reasonably protected from enemy action except by a direct hit.

(b) It is flexible hence can be rapidly re-arranged in the event of regrouping.

(c) Rapid in establishing communication. Works on the move although the range obtained will be much less than when stationary.

(d) Economical in personnel and equipment.

17. **Disadvantages of Net Radio**.

(a) Inherently insecure and susceptible to enemy interception which necessitates the use of codes and ciphers with a consequent delay in clearing traffic and an overall increase in operating personnel.

(b) Net Radio being inherently insecure demands a considerable degree of security consciousness on the part of the users. This means adherence to standard procedures and security codes.

18. **Radio Relay**. Radio relay implies that a series of radio transmitters and receivers are normally spaced between 20-35 Km apart and are used to provide point signal communication at enhanced ranges by the use of repeaters.

19. **Advantages of Radio Relay**.

(a) Replace line with considerable economy of manpower and stores.

(b) It can be operated over the area where for reasons of ground or enemy activity use of line may not be possible.

(c) Provides greater flexibility than line.

(d) Quick to set up and move except in mountainous country.

(e) Ability to employ multichannel equipment radio relay provides more data circuits over one link than can normally be provided over the average field cable.

(f) It has a much greater traffic handling capacity.

20. **Disadvantages of Radio Relay**.

(a) Liable to interception and hence insecure.

(b) Has relatively greater security than net radio, depending upon the siting and direction of the beams.

(c) Liable to interference from enemy jamming although not as much as in the case of net radio.

(d) Terrain between stations must be reasonably suitable to get a ‘quasi-optical, this presents difficulty in siting.

(e) The location of terminal and intermediate stations may not suit the tactical layout and may, therefore, create additional protection requirements.

(f) It needs critical siting and cannot work on the move.

(g) Slightly more expensive in men and material than in the case of net radio.

21. . Other more modern modes of communication are based on internet and Satellite Communication (SATCOM). These systems provide global communication capabilities, which are often used for secure, long-distance communication, which is done in digital formats. It does not require any physical connect within subscribers like line communication and has in built secrecy.

CONCLUSION

22. This is the basic means of signal communications for a static force. It is reliable, secure and practically free from electrical interference. Radio communication is flexible, works on move and can be established rapidly.

SUMMARY

- **Line**. This is the basic means of signal communications for a static force. It is reliable, secure and practically free from electrical interference. It is vulnerable to physical interference and enemy interception.
- **Radio Communication**. Radio communication involves Net Radio and Radio Relay. Net radio is the basic means of signal communication for any mobile forceand Radio communication is flexible, works on move and can be established rapidly. It is susceptible to enemy interception and requires use of codes, ciphers and standard procedures for voice communication. It is economical in terms of men and material than line communication.
- **Radio Relay**. Radio relay implies that a series of radio transmitters and receivers are normally spaced between 20-35 Km apart and are used to provide point to point signal communication. It is less liable to interference from enemy jamming in comparison to radio communication. Economical than line communication but more expensive than radio communication.

ASSESMENT EXERCISE

Multiple Choice Questions

Q1. Who invented the telephone?

(a) Issac Newton
(b) C.V. Raman
(c) Alexander Graham Bell
(d) Marie Curie

Q2. Which is a component of line communication?

(a) Telephone
(b) Trunk Media
(c) Local loop
(d) All of the above

Q3. How many types of waves are there?

(a) 2 (b) 3
(c) 4 (d) 5

Q4. How many types of propagation of electromagnetic waves are there?

(a) 2 (b) 3
(c) 4 (d) 5

Q5. Radio Communication involves?

(a) Net Radio
(b) Radio Relay
(c) Both A and B
(d) None of the above

Q6. Ionosphere is involved in which propagation of waves?

(a) Sky Wave
(b) Space Wave
(c) Ground Wave
(d) All of the above

Q7. Maximum communication distance can be achieved in which type of wave propagation?

(a) Sky Wave
(b) Space Wave
(c) Ground Wave
(d) Troposcatter

Q8. Minimum communication distance can be achieved in which type of wave propagation?

(a) Sky Wave
(b) Space Wave
(c) Ground Wave
(d) Troposcatter

Q9. The efficiency of Net Radio communication is affected by?

(a) Weather
(b) Terrain
(c) Power output of radio set
(d) All of the above

Q10. The efficiency of Net Radio communication is affected by?

(a) Weather
(b) Terrain
(c) Power output of radio set
(d) All of the above

Q11. There are how many types of Radio Communications?

(a) One
(b) Two
(c) Three
(d) Four

Q12. What does radio relay mean?

(a) Series of Radio Transmitters & Receivers
(b) Radio & Communication

Q13. Radio Relay implies a series of Radio Transmitters and Receivers which are normally spaced between which range?

(a) 10–15 km
(b) 20–35 km
(c) 15–20 km
(d) 40–50 km

Q14. What are the types of Radio Technology Communication?

(a) RT Conversation
(b) Formal Message
(c) Unregistered Message
(d) All of the above

Q15. Radio Communication involves the following:

(a) Open & Close
(b) Net Radio and Radio Relay
(c) Radio and Line
(d) Radio and Internet

Q16. What does radio relay mean?

(a) Series of Radio Transmitters & Receivers
(b) Radio & Communication
(c) Radio & Set
(d) Radio & Television

Short Answer Type Questions

Q1. What is Communication?

Q2. What are the components of line communication?

Q3. What are the basic components of Radio Communication?

Q4. What is Radio Communication?

Q5. What are the advantages of Line Communication?

Q6. What are the disadvantages of Radio Communication?

Long Answer Type Questions

Q1. What is Net Radio? What are the advantages and disadvantages of Net Radio?

Q2. What are the advantages and disadvantages of Line Communication?

Q3. What is Radio Relay? What are its advantages and disadvantages?

MILITARY COMMUNICATION (SD/SW)

CHAPTER MC II: BASICS OF RT PROCEDURE & PHONETIC ALPHABETS



TEACHING INSTRUCTIONS

Period : **04 (160 Mins)**
Type : **Lecture & Practice**
Year : **2nd Year (02) & 3rd Year (02)**
Conducting Officer : **Permanent Instructor**

Training Aids: Class Room, Computer with OHP, Screen, Pointer Staff, Presentation, Script/Book/Lesson Plan, Board and Markers.

Time Plan

- **Introduction : 05 Mins**
- **Part I : 10 Mins**
- **Part II : 10 Mins**
- **Part III : 20 Mins**
- **Part IV : 20 Mins**
- **Part V : 20 Mins**
- **Conclusion and Summary : 05 Mins**
- **Practice : 80 Mins**

INTRODUCTION

1. The RT procedure is laid out to communicate on the Radio to make the conversation secure and successful, which is referred to as “Radio Telephony”. This procedure is very important to avoid confusion over the radio and is generally used by defence forces, air traffic controllers, maritime operators, and amateur radio operators. A radiotelephone is a two-way radio ie. both sides can speak. Most frequencies are simplex i.e. only one person can speak at a time. When you press the PTT (Press to Transmit) button the receiving circuit is cut off. i.e. you can speak but you can't hear that is why when person A has finished speaking, he says “Over”, meaning over to you, before releasing the PTT. Person B now has the mike. The conversation then goes back and forth which each person passing the mike by saying “over” when he has finished. At the end of the conversation when one person has had enough, he says “Out”.



INTERESTING FACTS

- **Radiotelephony means telephony by radio**.
- **Radiotelegraphy is radio transmission of telegrams and Television is transmission of moving pictures and sounds**.

OLD WALKIE-TALKIE DESIGN



PREVIEW

The lecture will be conducted in the following parts:-

- **Part I: Basics of RT Procedures**
- **Part II: Phonetic Alphabets**
- **Part III: Characteristics of Walkie/Talkie**
- **Part IV: Standard Phrases**
- **Part V: Signal Strength**
- **Part VI: Practices of RT Procedure**

LEARNING OBJECTIVES

- **Understanding the basic principles of RT procedure,**
- **Advantages and disadvantages of RT communication**
- **Learning the universal phonetic alphabet**
- **Learn characteristics/features of GP 338 walkie talkie**
- **Learn basic control and handling of walkie-talkie**

PART I: BASICS OF RADIO TELEPHONY

2. The principles of Radio Telephony (R/T) in the Indian Army are focused on effective communication, ensuring clear and secure exchanges of information during operations. These principles guide the use of radio communication equipment in a disciplined and systematic manner to maintain operational readiness and coordination. It is a set of rules which is used to maintain security, speed and accuracy while sending news. It helps us to eliminate errors caused by different languages and dialects.

3. **Principles of Radio Telephony Procedure**. BASS defines the Principles of RT procedure. These include:-

(a) **B – Brevity**. All transmissions should be as short as possible so that the enemy cannot easily intercept them and time is not wasted in sending unnecessary messages. Avoid unnecessary words and ensure the message is easily understood. This is critical in the field, where time and clarity can be of the essence.

(b) **A – Accuracy**. Messages should be repeated back (read-back) to confirm they have been understood correctly. This ensures that no information is missed or misunderstood. If the meaning of the fixed correct words or digits/letters is known to both the speaker and the listener, then there will be fewer errors.

(c) **S – Security**. If the meaning of the fixed correct words or digits/letters is known to both the speaker and the listener, then there will be fewer errors. Sensitive information should be encrypted or transmitted in a secure manner to prevent interception by adversaries and operational security Radio operators should always maintain strict security protocols, such as avoiding the transmission of sensitive details over unsecured channels. If each unit uses its own specific modus operandi, the enemy will become familiar with their habits and will quickly recognize them.

(d) **S – Speed**. Radio equipment must be regularly tested and maintained to ensure it is operational at all times. This includes both hardware checks and frequency management to avoid interference. Stations must constantly monitor radio frequencies to receive any incoming messages or signals, ensuring no communication is missed during critical operations. By setting a system and rule for any work, that work can be done faster because it becomes easier for the speaker to say and the listener to understand.

4. In summary, the principles of Radio Telephony in the Indian Army revolve around clarity, security, discipline, and effectiveness in communication. These principles ensure that radio communication supports operational efficiency and helps maintain coordination and command in the field. In this section we will understand the definitions of the terms that are used in Radio Telephony Procedure (RT Procedure).

(a) **Radio Telephony Procedure (RT Procedure)**. It is a set of rules which is used to maintain security, speed and accuracy while sending news. It helps us to eliminate errors caused by different languages and dialects.

(b) **Code Sign and Link Sign**. To conceal the identity of the information giving and information receiving station, we use code sign and link sign.

(c) **Code Sign**. A group of three words which hides the identity of a unit/formation/headquarters. These are given by the above formation. For example, ABC, this can be the code sign of a battalion.

(d) **Link Sign**. It is made by adding numbers along with code sign, which hides the identity of the radio station. Like ABC 1- it can be the link sign of the company of the same battalion.

(e) **Appointment Code**. hides the rank of the officer/level of the commander. For example:- Battalion commander is also a tiger and section commander is also a tiger, 2IC of the Bn is known as Lamb.

(f) **Standard Phrases**: This is a group of words whose meaning is considered standard in the army. By using these, the sender understands the true meaning of the message without any delay.

(g) Proper management of radio frequencies is essential to avoid congestion and ensure effective communication. Military communications often use different frequency bands, and it's critical to stick to assigned frequencies to avoid interference with other units or external signals.

(h) Only authorized personnel should operate the radio equipment. Unauthorized transmissions should be avoided.

(j) Radio communication must also ensure proper coordination between different arms and units, such as infantry, artillery, and armored units, to execute coordinated operations effectively.

5. Various advantage/disadvantage RT procedure and radio communications are as under:-

(a) **Advantages**. It is easy to establish, flexible and enables transmissions to more than one station.

(b) **Disadvantages**. Easier to intercept, liable to atmospheric interference and interference from other stations, liable to be jammed and requires trained operators.

6. **Need for Standard Procedure**. Standard procedure in RT is needed to attain speed, uniformity, and security and prevent misunderstanding & confusion by use of code signs, link signs, etc. which hides the identity of an operator and that of a unit.

7. **Important Points to Remember During RT Communication**. RSVP is the key word which is to be kept in mind while making radio telephony communication:-

(a) R- Rhythm (b) S- Speed

(c) V- Volume (d) P- Pitch

8. **Types of RT Communication**.

(a) **RT Conversation**. This is the normal conversation between radio operators, which is not registered.

(b) **Unregistered (UR) Message**. The only record taken will be in the radio operator’s logs of the sender and receiver.

(c) **Formal Message**. It is written down and signed by the originator will be written on a message form (IAFU-4009).

PART II: PHONETIC ALPHABETS

9. Human beings with their higher intellect have successfully developed and evolved various forms of communications like, sign language, voice, written script, line transmission, radio wave, space wave and highly complex digital communication forms. The various means of communication adopted are as under:-

(a) **Voice Medium**. The development and graduation of sign to voice has also evolved, as voice is a very powerful means of communication, however this had its own disadvantages of limited reach. This was further restricted because of the loudness, pitch and ability of the speaker/ transmitter and the hearing ability of the receiver. It is also affected by other factors such as interference of surrounding sounds, noise pollution, echoing and attentiveness of the listener.

(b) **Written Script**. Once languages were formalised and some commonality in understanding of each other had been achieved, it appears that written script made a beginning. The earliest form of written expression can be found on the rock engravings of ancient scriptures in pictorial form. This gradually developed into alphabetic form and written script. Once the script had been evolved, messages could be communicated from the one person to another in written form.

(c) **Phonetic Alphabets**. Using a phonetic alphabet can increase communication accuracy and efficiency when there is a lot of background noise and when people speak different languages or dialects. The use of a phonetic alphabet (like Alpha for A, Bravo for B, etc.) helps avoid confusion over poor signal quality or noisy environments. The phonetic Alphabets are as under:-

PHONETIC ALPHABETS

A = ALFA	B = BRAVO	C = CHARLIE
D = DELTA	E = ECHO	F = FOXTROT
G = GOLF	H = HOTEL	I = INDIA
J = JULIETT	K = KILO	L = LIMA
M = MIKE	N = NOVEMBER	O = OSCAR
P = PAPA	Q = QUEBEC	R = ROMEO
S = SIERRA	T = TANGO	U = UNIFORM
V = VICTOR	W = WHISKY	X = X RAY
Y = YANKEE	Z = ZULU	

The concept of pronouncing of Signal codes are spoken by the sender such as "ALPHA", this means that receiver understands as English word "A".

PART III: CHARACTERISTICS OF WALKIE/TALKIE

10. **Definitions**

(a) **Radio Net**. Stations working on the same frequency to communicate with each other comprise a Radio Net.

(b) **Control Station**. Station serving as the senior HQ in the Radio Net. It is responsible for the establishment of communications and maintains radio discipline on the net.

(c) **Link Sign**. It is a secret group of letters or a combination of letters and figures, allotted to a station on the radio net, for concealing the identity of the communicating station. Link signs are changed daily or even earlier if required.

(d) **Code Sign**. It is a three-letter group allotted to HQ/formation or unit to conceal their identity. These are changed daily.

(e) **Long Message**. Message that lasts for more than half a minute generally consists of more than thirty groups.

(f) **Short Message**. A short message lasts for half a minute or less.

(g) **Standard Phrases**. The use of standard procedure ensures that the exact meaning of a sender is conveyed to the receiver in a minimum time.

(h) **Code Word**. A code word is a single word used to provide security cover for reference to a classified matter.

(j) **Nicknames**. A nickname consists of two separate words that do not bear any relationship to each other i.e. PAHAD BANDAR. Nicknames have very little security value and are used for convenience.

11. **Types of Calls**. Calls can be categorised as under:-

(a) **Single Call**. A call when only one station on a net is addressed by any station of the net.

(b) **Net Call**. A call is used to address all stations on a net e.g. “All stations Alpha”.

(c) **Multiple Calls**. A call when two or more but not all stations on the net are addressed. The pro word “and” will be inserted between the last two call signs e.g. `A1 for A2 and A3'.

(d) **Net Call with Exceptions**. A call used to address all stations on the net except a few. Here a net call is made and stations not called are specified e.g. “all stations Alpha except ABC 3”.

PART IV: STANDARD PHRASES

12. **Standard Phrases**. This is a group of words whose meaning is considered standard in the army. By using these, the sender understands the true meaning of the message without any delay. Some examples are as follows:-

(a) Hello - A way to draw attention to yourself.

(b) Correction - The last word of the message just sent is wrong, this is the correct one.

(c) SAY AGAIN - The receiver of the message asks for a repeat.

(d) I SAY AGAIN- I say it again.

(e) Roger - I have heard and understood the message.

(f) WILCO - The message has been heard, understood and implemented

(g) SO FAR - Reply to the messages sent so far, there are more messages to come.

(h) WAIT - Wait a few seconds.

(j) WAIT OUT - Wait for the next order.

PART V: SIGNAL STRENGTH AND TYPE OF CALLS

13. **Signal Strength**. Signal strength received during Radio Telephony conversations is classified as under:-

(a) **One**. The signal is very weak and not readable.

(b) **Two**. The signal is weak, and readable now and then.

(c) **Three**. The signal is fairly good and readable with difficulty.

(d) **Four**. The Signal is Good and readable.

(e) **Five**. The signal is very good and perfectly readable.

14. **Net Discipline**. Net discipline while making RT calls is most important for better and smoother communication. The following points must be followed to maintain the Net Discipline:-

(a) Follow the instructions of the Control Station.

(b) All Stations should respond in sequence.

(c) Give priority to the message.

(d) Without sequence do not answer

15. **Documentation**. Documentation forms a very important aspect of the RT procedure. All events need to be documented for future reference in case of a query.

16. **Rules on Security**. Security over radio is an important issue in defence forces hence there is a need to adopt these procedures and follow certain golden rules:-

(a) Think before you speak.

(b) Use the correct procedure.

(c) Avoid the use of jargon.

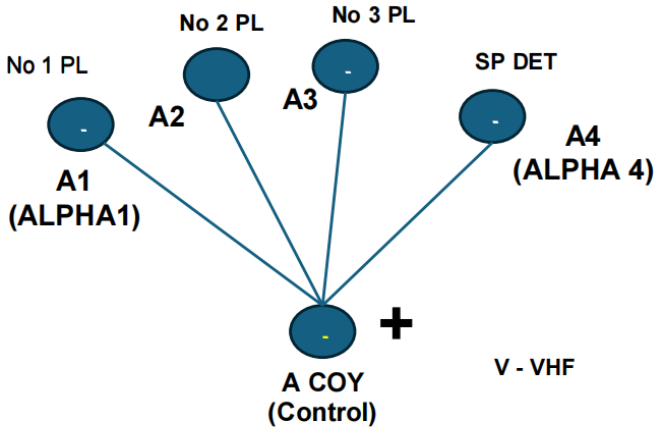
(d) Use official codes only.

(e) Substitute clear names with code signs.

(f) Be brief.

PART VI: PRACTICE OF RT PROCEDURE

17. **Basic Radio Net Diagram with Link Signs**.



18. **Basic Conversations**.

(a) **Control Station (+):** All Stations Alpha Report **Over**

(b) A1 Outstation: Alpha 1 Ok **Over**

(c) A2 Outstation: Alpha 2 Ok **Over**

(d) A3 Outstation: Alpha 3 Ok **Over**

(e) A4 Outstation: Alpha 4 Ok **Over**

(f) **Control Station(+):** All Stations Alpha Report Your Signal Strength **Over**

(g) A1 Outstation: Alpha 1 Strength 4 **Over**

(h) A2 Outstation: Alpha 2 Strength 2 **Over**

(i) 3 Outstation: Alpha 3 Strength 5 **Over**

(k) A4 Outstation: Alpha 4 Strength 3 **Over**

(l) C**ontrol Station (+):** All Stations Alpha By 1400h Send Your Ammunition Demand **Over**

(m) A1 Outstation: Alpha 1 Roger **Over**

(n) A2 Outstation: Alpha 2 Roger **Over**

(o) A3 Outstation: Alpha 3 Roger **Over**

(p) A4 Outstation: Alpha 4 Roger **Over**

(q) **Control Station (+):** All Stations Alpha Report Your Troop Strength As On Date And Forward Your Kerosine Demand For Next Month By 1700h **Over**

® A1 Outstation: Alpha 1 Wilco **Over**

(s) A2 Outstation: Alpha 2 Wilco **Over**

(t) A3 Outstation: Alpha 3 Wilco **Over**

(u) A4 Outstation: Alpha 4 Wilco **Over**

(v) **Control Station (+):** All Stations Alpha Nothing Else **Out**

CONCLUSION

19. Radio Telephony" procedure is very important to avoid utter confusion over the radio and is generally used by defence forces, air traffic controllers, maritime operators, and amateur radio operators. A radiotelephone is a two-way radio ie. both sides can speak. Most frequencies are simplex ie only one person can speak at a time.

SUMMARY

- **Radio Telephony**. The procedure laid out to communicate on the Radio so as to make the conversation secure and successful, which is referred to as "Radio Telephony". The Principles of Radio Telephony Procedure are:-
  - B – Bravity
  - A – Accuracy
  - S – Security
  - S – Speed
- A walkie-talkie is a hand-held, portable, two-way radio transceiver which operates in VHF and UHF mode.
- **Facilities/ Features**. The following are the facilities/features in handheld Radio set GP338:-
  - It is portable and light in weight. It can be operated easily and so operated in VHF/UHF and 2-way simplex mode. It has 128 channels of this radio set can be preset into 8 zones. Option of selective call facilities available.
  - A call alert can be given to the receiver station. It can select the required zone.
  - In addition, Start/stop facilities of scan operation available to Add/Delete from scan list facilities available.
- **Communication Range**. The range radio set with repeaters it can work from Principles of Radio Telephony Procedure.
  - Without repeater – 4 to 5 km
  - With repeater – 20 to 40 km
- **Power Output**. It has Power output (HF) of 1 to 5 watts and Power output (UHF) 1 to 4 watts.

ASSESMENT EXERCISE

Multiple Choice Questions

Q1. What is the full form of RT?

(a) Radio Transmitter
(b) Radio Transfer
(c) Radio Term
(d) Radio Telephony

Q2. The keyword for principles of Radio Telephony is known as?

(a) BAAS
(b) BASS
(c) BOSS
(d) BAUS

Q3. The keyword for Radio Telephony Procedure is known as?

(a) RSPV
(b) RSVP
(c) VPRS
(d) Non of above

Q4. How many phonetic alphabets are there?

(a) 22
(b) 26
(c) 20
(d) 28

Q5. How many types of calls are there?

(a) One
(b) Two
(c) Three
(d) Four

Q6. A secret group of letters and figures allotted to a station in the radio net is called?

(a) Code Sign
(b) Link Sign
(c) Code Word
(d) Nickname

Q7. Letter group allotted to an HQ/Formation/Unit to conceal identity in the radio net is called?

(a) Link Sign (b) Code Sign
(c) Nickname (d) Code Word

Q8. A single word used to provide security cover for reference to a classified matter is called?

(a) Link Sign (b) Code Sign
(c) Code Word (d) Nickname

Q9. Code Sign and Link Sign are used in radio procedures to:

(a) Conceal the identity of the information
(b) Conceal the name of the instrument used for communication
(c) Describe the name of the sender
(d) Facilitate the enemy about the message

Q10. Two separate words (e.g., *PAHAD BANDAR*) that don't bear any relationship and have very little security value are called?

(a) Link Sign
(b) Code Sign
(c) Code Word
(d) Nickname

Q11. What is the standard phrases used in RT procedures?

(a) Over
(b) Roger
(c) Wilco
(d) All of the above

Q12. A message that lasts for half a minute or less is called:

(a) Long Message
(b) Short Message
(c) Medium Message
(d) Both (b) and (c)

Q13. What is the appointment code of a Commander?

(a) Eagle
(b) Sheep
(c) Tiger
(d) Lion

Q14. What phrase used in RT procedure for "My transmission has ended and I do not expect to hear a further transmission from you on this subject" is called:

(a) Out
(b) Over
(c) Over and Out
(d) Wait Out

Q15. Phrase used in RT procedure for “Message received and understood” is called:

(a) Ok
(b) Understood
(c) Roger
(d) Wilco

Short Answer Questions

Q1. What are the rules used for security on the radio?

Q2. What is radio telephony procedure?

Q3. What is Radio Telephony? Write the advantages of RT.

Q4. What is advantage of appointment code?

Q5. What are the types of Calls? Define each of the following types.

Q6. What is Call?

Q7. How many types of Call?

Long Answer Questions

Q1. What is the principles of Radio Telephony Procedure?

Q2. What are the standard phrases used in the RT Procedure?

Q3. What are the important points to remember during Radio Telephony?

MILITARY COMMUNICATION (SD/SW)

CHAPTER MC III: WIRELESS COMMUNICATION & LATEST TRENDS IN COMMUNICATION

"Soldiers can solve 98 percent of their problems by just talking to someone about them. All you have to do is listen."(SMA William G. Bainbridge).



TEACHING INSTRUCTIONS

Period	**:**	**01 (40 Mins)**
Type	**:**	**Lecture**
Year	**:**	**2nd Year SD/SW**
Conducting Officer	**:**	**Permanent Instructor**

Training Aids: Class Room, Computer with OHP, Screen, Pointer Staff, Presentation, Script or Book Flagged or Lesson Plan in File, Board, and Markers.

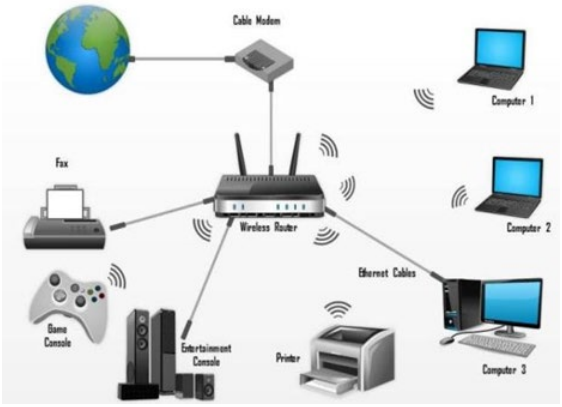
Time Plan

➢ **Introduction**	**:**	**03 Mins**
➢ **Part I**	**:**	**10 Mins**
➢ **Part II**	**:**	**10 Mins**
➢ **Part – III**	**:**	**15 Mins**
➢ **Conclusion and Practice**	**:**	**02 Mins**

NTRODUCTION

INTERESTING FACTS

- **Wi-Fi stands for Wireless Fidelity which is a wireless technology standard for wireless Internet access**.
- **A Wi-Fi network is simply an internet connection shared with multiple devices in a home or business via a wireless router. The router is connected directly to your internet modem and acts as a hub to broadcast the internet signal to all your Wi-Fi-enabled devices**.



HOME WIRELESS NETWORK DIAGRAM

PREVIEW

The lecture will be conducted in the following parts:-

- **Part I: Wireless Communication**
- **Features and Benefits.**
- **Part II: Terminal Equipment and Limitations of Wi-Fi.**
- **Part III: Latest trends and development in communication**.

LEARNING OBJECTIVES

- **Learn about Wireless communication, its features and benefits**.
- **Learn terminal equipment being used for Wi-Fi**.
- **Understanding limitations of Wi-Fi technology**.
- **Learn latest trends and development in communication**.

PART I: WIRELESS COMMUNICATION

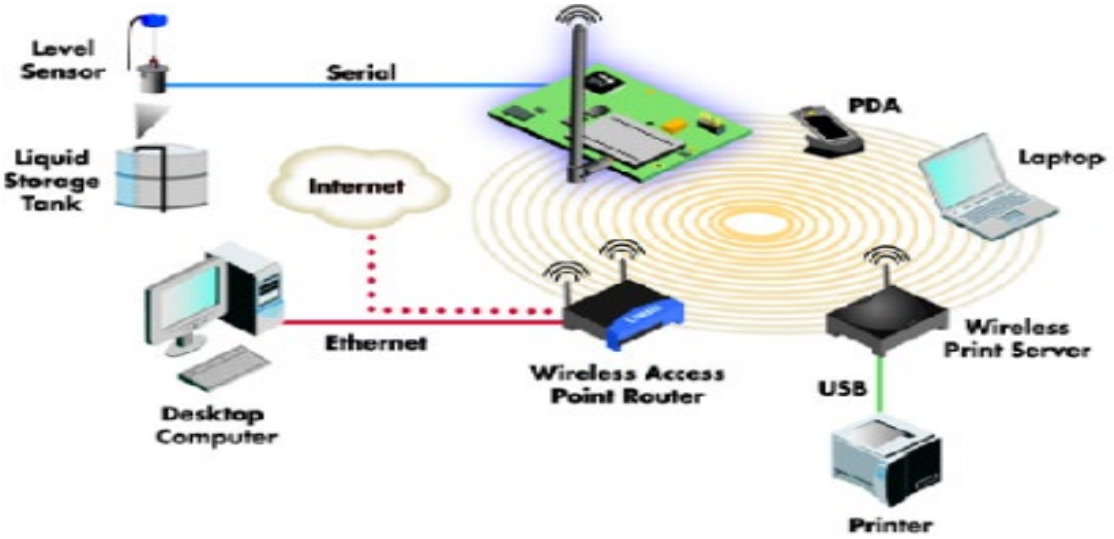
1. Wireless communication provides its users with the liberty of connecting to the internet from any place such as their home, office, or a public place without the hassles of plugging in the wires. It is quicker than the conventional modem for accessing information over a large network. Wi-Fi uses a radio network to transmit data between its nodes. Such networks are made up of cells that provide coverage across the network. The more the number of cells, the greater and stronger the coverage on the radio network.

Features and Benefits of Wireless Technology

2. Wireless communication operations permit services, such as long-range communications, which are impossible or impractical to implement with the use of wires. Information is transferred in this manner over both short and long distances. The following list summarizes some of the benefits of a Wireless network:-

(a) **Unmatched Mobility and Elasticity**. Wireless communication allowing the new intensity of connectivity without giving up functions. Wi-Fi introduced various types of utilities such as music streamers that transmit your music to speakers without

any wire you can also play music from the remote computer or any other attached to the network. The Wi-Fi technology system is remarkable, as a user can download songs, send an email and transfer files expediently at phenomenal speed even whilst the user is on the move.



(b) **Fortress Technology**. Wi-Fi supports secure wireless solutions and ensures the privacy of the user's data.

(c) **Support an Entire Age Bracket**. Wi-Fi technology has several advantages, it supports an entire age bracket creates a connection between components on the same network can transfer data between devices, and enables different kinds of devices such as games, MP3 players, PDA, and much more.

(d) **Convenient and Available Everywhere**. Wi-Fi is a convenient technology and where the network range station exists you are online during travel. You can equip yourself with a Wi-Fi network and get connected. You will automatically connect to the internet if you are near a hotspot. These days Wi-Fi exists everywhere with all its wonders.

(e) **Faster and Secure**. With Wi-Fi, you can get a high speed of internet because it is much faster than a cable connection you can establish a Wi-Fi network in a small space now you don't need any professional installation just connect to a power outlet with an Ethernet cord, and start browsing. Wi-Fi security system for Threats makes it more renewable and its tool protects your VPN and secure web page. You can easily configure the device to take better performance. The standard devices, embedded systems, and network security make it more powerful.

(f) **Wi-Fi with no limitation**. You can use a "Wi-Fi" network with no limitations because it can connect you worldwide. You can easily cater to your requirements with Wi-Fi networking applications because the power consumption is not very high compared to other bandwidth networks. The future of wireless internet network communications is bright.

(g) **Extension of Wi-Fi Technology**. It is because of this wireless technology that so many other advancements could take place. Have you ever thought of your TV to be supported by Wi-Fi? Well, if you didn't, start thinking now. There are smart TVs

in the market that connect to the internet. Having the internet on TV makes it possible to watch YouTube videos, Netflix and so much more. Read more about Wi-Fi TV technology on the World Wide Web.

(h) **Cost Reduction**. As mentioned above, the absence of wires and cables brings down costs.

(j) **Flexibility**. Extended access, cost reductions, and mobility create opportunities for new applications as well as the possibility of creating new solutions for legacy applications.

PART II: TERMINAL EQUIPMENT AND LIMITATIONS OF WI-FI

3. Emerging communication trends and issues are the new or evolving developments and challenges that affect the communication environment, such as technology, media, culture, politics, and society. Wi-Fi technology may not be efficient to perform the commutations. while wireless transmission media offer many advantages, they also have several limitations. These include interference from other devices and physical obstacles, security risks, limited range, and lower speeds compared to wired connections.

4. **Mobile Telephones**. One of the best-known examples of wireless technology is the mobile phone, also known as a cellular phone, with more than 8.9 billion mobile cellular subscriptions worldwide as of the end of 2024. These wireless phones use radio waves from signal-transmission 2025 towers to enable their users to make phone calls from many locations worldwide.



5. **Wireless Data Communications**. Wireless data communications are an essential component of mobile computing. The various available technologies differ in local availability, coverage range, and performance, and in some circumstances, users must be able to employ multiple connection types and switch between them.

6. Wi-Fi technology is not perfect and has many flaws that limit its use as follows:-

(a) **Security** Because wireless transmissions can pass through walls, security is an issue.

(b) **Wireless Reception**. Varies from area to area, even within your apartment. It's not always guaranteed that you'll have a connection to the Internet.

(c) **Interference**. Call quality is greatly influenced by the environment, and is particularly sensitive to electromagnetic radiation generated by other household appliances.

(d) **Compatibility Issue**. Despite global standardization, many devices from different manufacturers are not fully compatible, which in turn affects the speed of communication.

DID YOU KNOW?

- India's communication satellites, primarily under the GSAT and INSAT series play a vital role in communication infrastructure.

PART III: LATEST TRENDS AND DEVELOPMENT IN COMMUNICATION

7. **Troposcatter, Modem, Fax, Data Troposcatter**. The lower layer of the atmosphere below 15Km height is called a tropospheric region. Communication carried out in this layer uses the principle of troposcatter. In this system, microwaves are transmitted in the Ultra High Frequency (UHF) and Super High Frequency (SHF) bands to achieve Radio Communication over the horizon covering a range between 70 Km to 1000 Km.



8. **MODEM (Modulator-Demodulator)**. This device is used to convert computer-generated output (Digital signals) that can be transmitted on a telephone line. Modems are required at both the sending and receiving computers.



9. **FAX**. This is the common short form of FACSIMILE which is one of the memory-type electronic mail and message systems with the following advantages:-



(a) Can transmit graphics as well as Alphanumeric information (letters and numbers).

(b) Reduce time and eliminate transmission error.

(c) Use any transmission medium e.g. Telephone, line, micro radio wave.

10. **Satellite**. An object that revolves around another larger object whose motion is primarily and permanently determined by the force of attraction of the body is known as a satellite. Satellites are used for many purposes. Common types include military and civilian Earth observation satellites, communications satellites, navigation satellites, weather satellites, and space telescopes. Space stations and human spacecraft in orbit are also satellites. Satellite orbits vary greatly, depending on the purpose of the satellite, and are classified in several ways. Well-known (overlapping) classes include low Earth orbit, polar orbit, and geostationary orbit.

11. **Types of Satellite**.



(a) Weather Satellite.

(b) Scientific Satellite.

(c) Communication Satellite.

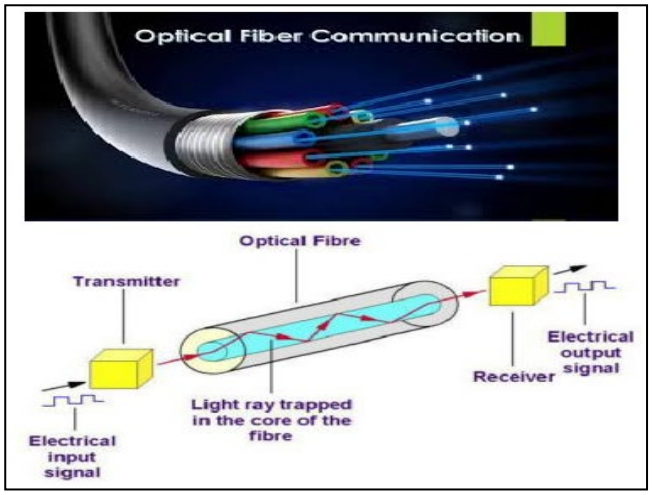
(d) Navigational Satellite

(e) Military Satellite.

12. **Communication through Optical Fibre Cable**. Hollow tubes made of corning glass with an outer protective coating of rubber/plastic etc. are what constitute optical fibers. These fibers are very delicate and small in diameter. The advantages and disadvantages are as under:-

(a) **Advantages**. It has a wide bandwidth carrying different types of information from low-speed voice signals to high-speed computer data. Less power requirement, small cable size and no electromagnetic interference.

(b) **Disadvantages**. It is expensive in terms of equipment and manufacturing. Requirement of experts to run the system always needed.



13. **Cell Phone**. The Cellular radio network was first introduced in Tokyo in 1979. It provides mobile subscriber access to the global telephone network. It is a rapidly expanding technology with high rates of obsolescence. The advantages and disadvantages are as under:-

(a) **Advantages**. Cell phone has more subscribers and traffic capability. There is no perceptible difference between mobile and fixed subscribers. It has better quality of service, higher speed of data exchange and can be used in an integrated mode with the computer network.

(b) **Disadvantages**. The cell phone is a open source. It can be jammed if required. Repeated charging of battery is required, it has many security-related issues so works on the capability and commercial interests of the service provider. It has health hazards due to excessive radiation.

14. **Internet**. Millions of computers all over the world are interlinked through telephone lines, satellites, submarine cables, and optical fiber networks. This World Wide Web (www) is what is called the “Internet”. It provides an instant, trouble-free, and cheap means of communication. The Internet is therefore a collection of individual data networks connected in such a way that data can be exchanged back and forth between networks widely separated. Electronic Mail, web browsing, and Voice Mail are the main facilities of the Internet.

15. **Multimedia**. It is a computer technology that displays information using a combination of full-motion video animation, sound graphics, and text with a high degree of user interaction.

16. **Video-Conferencing Systems**. These provide the full benefits of face-to-face communication with sound, graphics, and simultaneous transmission of data. The system enables people widely separated geographically to interact without having to meet in one place.



17. **Videophone**. It is a system that enables us to transmit an image via a digital telephone network, making visual contact has been made possible over great distances, apart from transferring speech. Facilities provided by videophone are:-



(a) Can transmit speech as well as colour video.
(b) Conduct of video conferences.
(c) Called subscriber is seen on the monitor.
(d) High quality of voice.
(e) Speed of sending/ receiving can be adjusted by the user.
(f) Map overlays can be transmitted.

CONCLUSION

18. Today technology has advanced in all spheres of life. Even the defence forces are well poised to exploit the state-of-the-art modern communication techniques to meet the requirements of the Indian Army in the 21st century. Some of the areas that the army is already in the process of exploiting are Cellular Radio (both GSM & CDMA), WLL, mobile trunked radio, mobile satellite systems, OFC, and so on. Therefore, it is very important to be well-versed with the latest trends in information technology to gain maximum advantages from the same.

SUMMARY

- **Wireless**. Wireless operations permit services, such as long-range communications, that are impossible or impractical to implement with the use of wires.

- **Mobile Telephones**. These wireless phones use radio waves from signal-transmission towers to enable their users to make phone calls from many locations worldwide.
- **Wireless Data Communications**. Wireless data communications are an essential component of mobile computing. The various available technologies differ in local availability, coverage range, and performance.
- **Wi-Fi**. Wi-Fi is a wireless local area network that enables portable computing devices to connect easily to the Internet. Wi-Fi has become the de facto standard for access in private homes, within offices, and at public hotspots. Cellular data service offers coverage within a range of 10-15 miles from the nearest cell site.

- Troposcatter is a system in which microwaves are transmitted in the UHF and SHF band to achieve radio communication over the horizon covering a range between 70 Km to 1000 Km.
- A modem (modulator-demodulator) is a network hardware device that modulates one or more carrier wave signals to encode digital information for transmission and demodulates signals to decode the transmitted information.
- A satellite is an artificial object which has been intentionally placed into orbit. Such object is sometimes called artificial satellites to distinguish them from natural satellites such as Earth's Moon.
- Satellites are used for a large number of purposes. Common types include military and civilian earth observation satellites, communications satellites, navigation satellites, weather satellites, and research satellites.
- Multimedia is content that uses a combination of different content forms such as text, audio, images, animation, video, and interactive content. Multimedia contrasts with media that use only rudimentary computer displays such as text-only or traditional forms of printed or hand-produced material.

ASSESMENT EXERCISE

Multiple Choice Question

Q1. Important features of Wi-Fi involve which undermentioned option?

(a) Mobility
(b) Fortress Technology
(c) Cost Reduction
(d) All of the Above

Q2. Limitations of Wi-Fi Technology is which undermentioned option?

(a) Security
(b) Interference
(c) Wireless Reception
(d) All of the Above

Q3. The lower layer of the atmosphere below ___ km is called the tropospheric region.

(a) 15 km
(b) 48 km
(c) 64 km
(d) 30 km

Q4. What advantages does Optical Fibre Communication offer?

(a) High Speed
(b) Less Power Requirement
(c) No Interference
(d) All of the Above

Q5. The cellular radio network was first launched in which year?

(a) 1979
(b) 1980
(c) 1983
(d) 1985

Q6. Which of the following is not a communication satellite?

(a) CARTOSAT-3
(b) GSAT-10
(c) GSAT-7C
(d) INSAT-3A

Q7. Which of the following is not part of Information Technology?

(a) Computer (b) Internet
(c) Telephone (d) Email

Q8. MODEM is used for transmitting digital signals over?

(a) Radio Set (b) Email
(c) Mobile Phone (d) Telephone Lines

Q9. Which of the transmission media cannot be used by FAX?

(a) Microwave Radio (b) Telephone
(c) Line (d) Hand Held Radio Set

Q10. Which application can be used for video conferencing?

(a) Google Meet
(b) Cisco Webex
(c) Zoom
(d) All of the Above

Q11. What is the benefit of Wi-Fi?

(a) Power Saving
(b) Cost Less
(c) Mobile Data
(d) Small Set

Q12. Video Conferencing can be held on _____?

(a) Motorola
(b) Telephone
(c) Wi-Fi Network
(d) Radio Set

Q13. This word 'www' is what is called the “Internet”.

(a) World Wide Web
(b) World War
(c) Wide Area Web
(d) Work Web for Net

Q14. Wi-Fi Network is ……?

(a) Slow and secure
(b) Faster and secure
(c) Limited to LAN
(d) Need Radio Set

Q15. Video Conferencing is useful for _____?

(a) Distance Talk
(b) Office Complex
(c) Wi-Fi Network
(d) Radio Set

Short Answer Questions

Q1. What is MODEM?

Q2. What is the Internet?

Q3. What is Tropo Scatter?

Q4. What is Fax?

Q5. What is Satellite?

Q6. What is Information Technology?

Q7. What is Multimedia?

Long Answer Questions

Q1. What is Wi-Fi? Write the benefits of a Wi-Fi Network.

Q2. What is Optical Fibre? Write the advantages of Optical fiber Cable

Q3. What are the features of Wi-Fi Technology?

Q4. What are the advantages of video phones?

MILITARY WEAPONS & EQUIPMENTS

5

CHAPTER WISE INDEX: MILITARY WEAPON & EQUIPMENT (JD/JW)

SER NO	SUBJECT CODE AND CONTENT	PAGE NO
CHAPTER MWE I: ORGANISATION OF AN INFANTRY BATTALION AND ITS WEAPONS		
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MILITARY WEAPON & EQUIPMENTS (SD/SW)

CHAPTER MWE-I: ORGANISATION OF AN INFANTRY BATTALION AND ITS WEAPONS

"The Infantry doesn't claim to have done it all, but without them, nothing else gets done."

"युद्धाय कृतनिश्चयः।"



TEACHING INSTRUCTIONS

Period	:	**03 (120 Mins)**
Type	:	**Lecture and Practice**
Year	:	**1st Year (02) & 3rd Year (01)**
Conducting Officer	:	**Permanent Instructor**

Training Aids: Class room, OHP, Board and Chalk/Markers

Time Plan

- **Introduction : 05 Mins**
- **Part I : 35 Mins**
- **Part II : 35 Mins**
- **Conclusion and Summary : 05 Mins**
- **Visit to Infantry Battalion : 40 Mins (Practical)**

INTRODUCTION

1. The Infantry Battalion stands as the cornerstone of the Indian Army, embodying resilience, versatility, and unyielding commitment to safeguarding the nation's sovereignty. As the primary combat force, the Infantry Battalion is meticulously trained and equipped to operate effectively under the most challenging and adverse conditions, demonstrating unmatched adaptability and endurance. Uniquely capable of fighting both independently and as part of a larger force, the Infantry Battalion's operational versatility makes it indispensable in modern warfare. Whether executing offensive operations, holding defensive positions, or undertaking specialized missions, the battalion is designed to persevere and succeed, even against formidable adversaries. Its inherent sustenance power, coupled with the unwavering motivation of its personnel, enables it to maintain combat effectiveness and morale through protracted engagements.

2. The Infantry Battalion's balanced force structure equips it to adapt seamlessly to diverse combat scenarios. In offensive operations, it can breach fortified enemy positions and dominate the battlefield. During defensive operations, it excels in holding critical terrain and repelling enemy advances. Moreover, its proficiency in special operations ensures it can execute precision missions that demand stealth, speed, and tactical excellence. The battalion's combat capabilities are further enhanced by its integral support weapons, which provide decisive firepower. These support weapons, including the 81mm Mortar and the Anti-Tank Guided Missile (ATGM) form the backbone of its firepower at extended ranges.

3. In addition to these weapons, the Infantry Battalion integrates its firepower with the tactical use of fortified positions, mobility, and synchronized maneuvers. These attributes ensure that the battalion can effectively disrupt enemy plans, cause maximum damage, and dominate the battlefield across diverse operational theaters. Ultimately, the Infantry Battalion exemplifies the ethos of the Indian Army - a balanced, determined, and relentless force, prepared to defend the nation and achieve victory under any circumstances.

INTERESTING FACTS

- **Largest & Oldest Combat Arm**.
- **Largest Contributor to the United Nations**.
- **Specialises In High Altitude Warfare**.
- **Queen of the Battle**.
- **Highest Param Vir Chakras**.



PREVIEW

- **Part I: Infantry Battalion Organisation**
- **Part II: Capability & Basic Weapons**.

LEARNING OBJECTIVES

- **Organization of Infantry Battalion & Its Constituents**
- **Capability & Employability**

PART I: ORGANISATION OF INFANTRY BATTALION

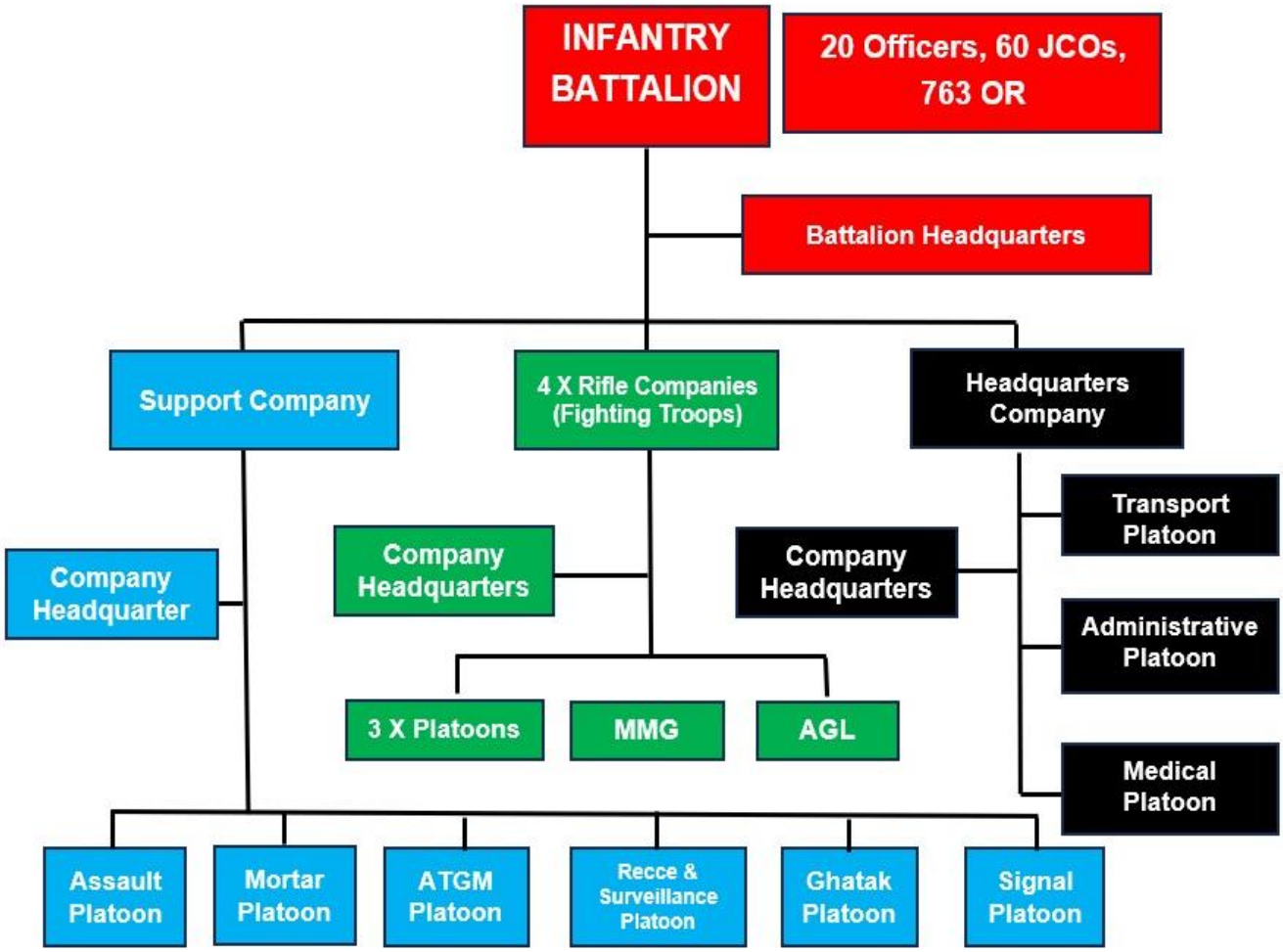
4. The Infantry battalion is the most important organisation of the Indian Army. It is trained and equipped to face any threats and challenges. It has the capability to attack/defend and hold ground as per task assigned. It can fight an enemy independently or as part of a larger force. It has the sustenance power and motivation to fight till last man last round. The Infantry battalion has **Four Rifle Companies**, each Company has **Three Platoons** and Each Platoon has **Three Sections**.



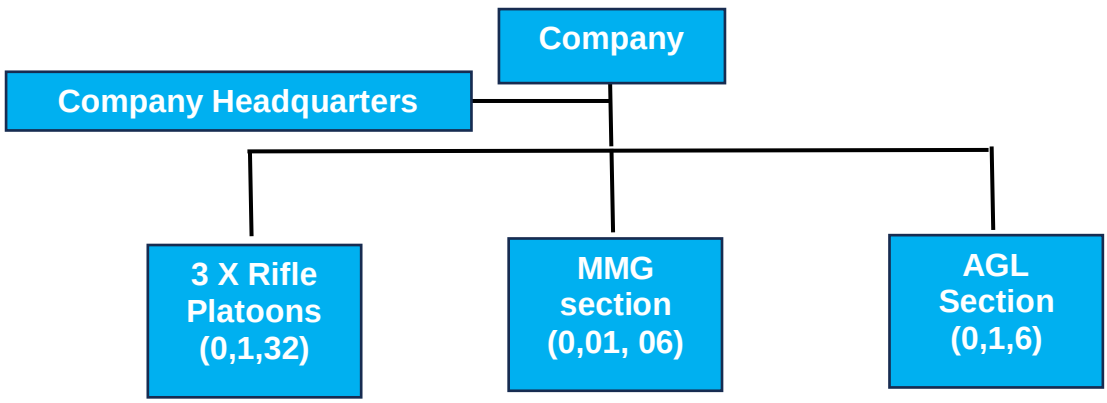
5. To Support the rifle companies to fight there is a requirement of various Support organisations within the battalion like administrative support, communication, recce and surveillance etc. To keep these support organisations at one place there is **One Support Company** within the battalion. Also to keep command and Control elements at one place, there is another company called as **Headquarter Company**.



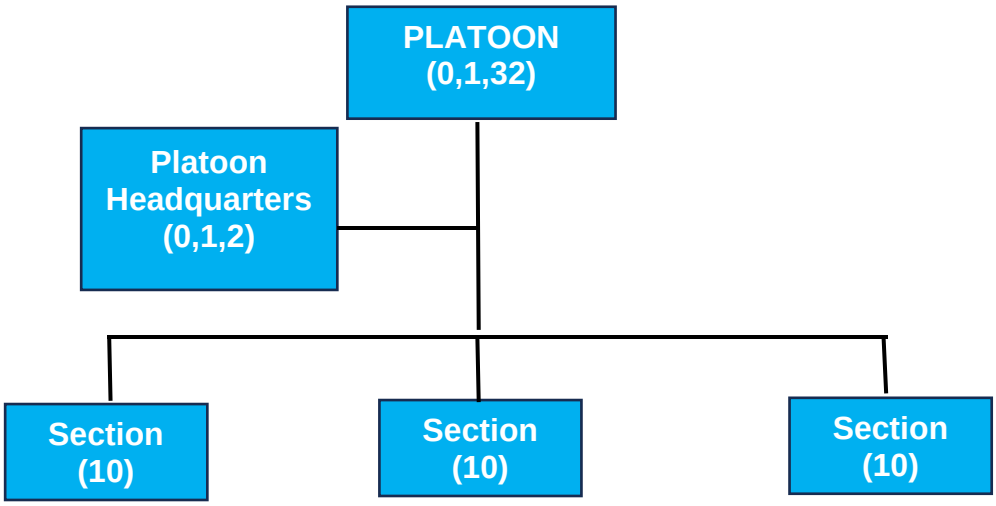
ORGANISATION OF INFANTRY BATTALION



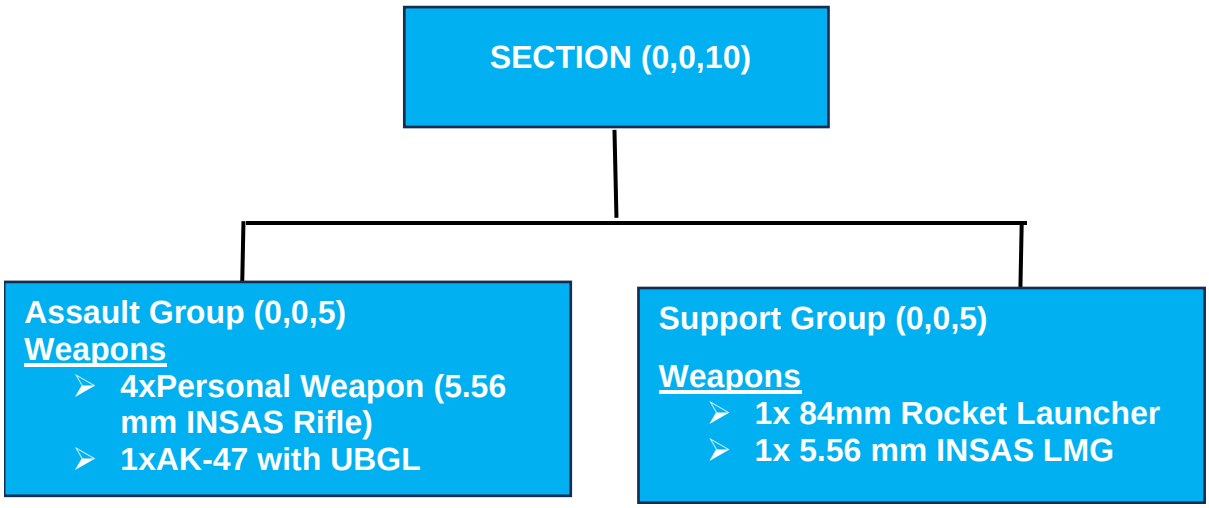
02 OFFICERS, 07 JCOs AND 117 OR (HAV, NKs & SEP)



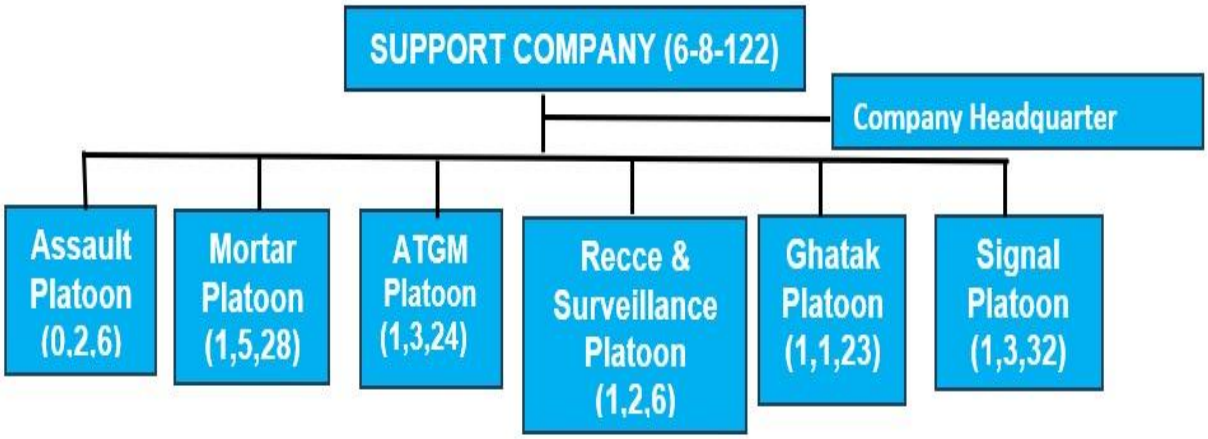
6. **Rifle Platoon**. The Rifle Platoon plays a crucial role in the combat operations of an Infantry Battalion's Rifle Company. It brings together soldiers who have undergone rigorous training and maintains strict discipline, enabling them to carry out tactical missions in a wide range of settings. The platoon comes equipped with light firearms and backup weaponry making it ready to take on offensive and defensive tasks as well as scouting duties. Led by a Junior Commissioned Officer, the Platoon serves as the core of the battalion's fighting power.



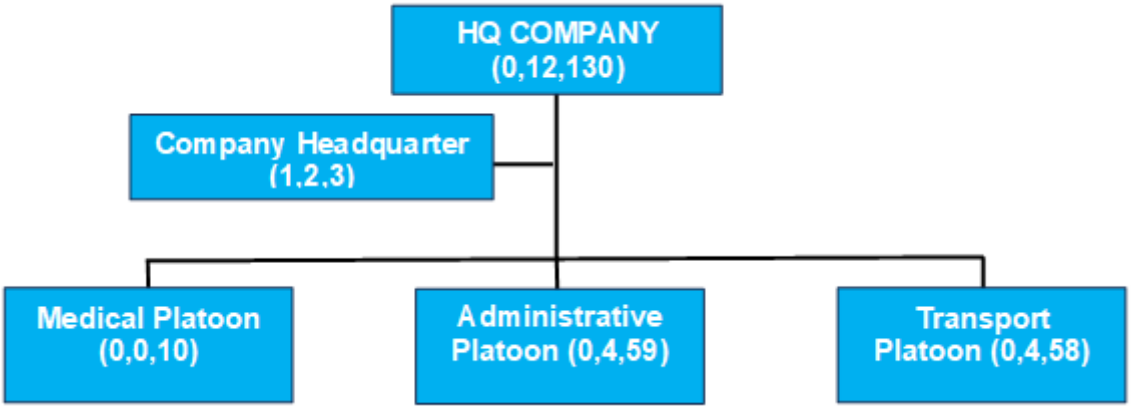
7. **Rifle Section**. Rifle Section is the smallest combat element of an Infantry Battalion consisting of 10 soldiers. This is the basic fighting unit which can operate independently in hand to hand combat situations. Commanded by a Non Commissioned Officer, it is a well-coordinated and swift team that employs small arms and support weapons in the pursuit of tactical objectives.



8. **Support Company**. The Support Company of an Infantry Battalion is vital in accomplishing operational success of the battalion through provision of the required support in firepower, reconnaissance, and communication. It consists of various Platoons, all of whom are trained to meet needs in the battlefield. The ATGM Platoon is Anti Tank Guided Missile Platoon capable of destroying enemy tanks by firing wire guided Missiles.



9. **HQ Coy**. The Headquarter (HQ) Company of an Indian Army Infantry Battalion serves as the command-and-control center, ensuring seamless coordination and execution of operations. It is the backbone of the battalion, providing logistical, administrative, and operational support to all other companies.



10. **Capabilities**. The capabilities of Infantry Battalion are as under:-

(a) **Self Reliance**. The trained soldiers are ready to independently engage with their opponents without being aided in combat by any other forces as the Infantry possesses all the necessary deterrents.

(b) **Ability to Hold Ground**. An Infantry battalion is competent enough to stand their ground with or without external assistance.

(c) **Adaptability**. An Infantry battalion can perform on all the types of terrains during all the time, even under difficult climatic conditions. The Infantry battalion can be readily transported by land, sea, or air to the battle area.

(d) **Mobility**. Infantry battalion gets the greatest mobility. It can go almost anywhere and everything including nearly all barriers.

(e) **Vulnerability**. An Infantry battalion is in jeopardy of being assaulted with devastating attacks from any charging tanks, artillery fire, small weapon artillery, aerial bombardments and anti-personnel mines.

11. **Employability**. At a fundamental level, an Infantry battalion is tasked with physically engaging enemy forces while seizing or defending territory. The use of firepower in conjunction with movement forms the basis of Infantry tactics. This understanding is fundamental for any Infantry battalion starting from a section level upwards. An Infantry battalion along with its sub-units is trained in accomplishing missions against the enemy in a wartime area without the need for complete reliance on support by other arms but through effective combination of terrain, weapon systems, and above all the Infantry men with their determination, courage, and desperation.

PART II: BASIC WEAPONS OF INFANTRY BATTALION

12. **Personal, Company/Battalion Support Weapons**. Personal/basic weapons of a soldier of Infantry Battalion consists of weapons such as 5.56mm INSAS Rifle, 7.62mm AK-47 along with Under Barrel Grenade Launcher, Carbine, 9mm Pistol, Number 36 Hand Grenade, 84mm Rocket Launcher, 7.62mm Sniper Dragunov and 5.56mm INSAS LMG. Apart from this there are company level support weapons like 30mm Automatic Grenade Launcher (AGL) and 7.62mm Medium Machine Gun (MMG). Battalion Support Weapons like 81mm Mortar and Anti-Tank Guided Missile (ATGM) are held by Support Company and are used on orders from the Battalion Commander. Details are given in tabulated form below.

5.56MM INSAS RIFLE

SPECIFICATION/ FEATURES

Manufacturer	Indian Ordnance Factory Board
Weight	3.69 kg
Calibre	5.56 mm
Length of Rifle	960 mm
Effective range	400 metres
Principle of Operation	Gas Operated
Mode of fire	Single Shot, Three Round Burst (TRB)
Rate of Fire	
Normal	60 rounds/minute
TRB	90 rounds/minute
Intense	150 rounds/minute
Cyclic	600-650 rounds/minute



5.56MM INSAS LMG (LIGHT MACHINE GUN)

SPECIFICATION/ FEATURES

Manufacturer	Indian Ordnance Factory Board
Weight with Bipod	7.2 kg
Calibre	5.56 mm
Length with Bipod	1030 mm
Effective range	700 metres
Principle of Operation	Gas Operated, Rotating Bolt Mechanism
Mode of fire	Single Shot and Automatic
Rate of Fire	
Normal	60 rounds/minute
Intense	120 rounds/minute
Cyclic	600 rounds/minute



7.62 MM DRAGUNOV SNIPER RIFLE

SPECIFICATION/ FEATURES

Manufacturer	Soviet Union
Weight with Bipod	4.3 kg
Calibre	7.62 mm
Length with Bipod	1030 mm
Effective range	800 metres
Range with telescope sight	1300 meter
Magazine capacity	10 rounds
Ammunition Fired	• Armor Piercing. • Sniper Balls. • Steel Core. • Tracer. • Incendiary.



84 MM RL (ROCKET LAUNCHER)

SPECIFICATION/ FEATURES

Manufacturer	Indian Ordnance Factory Board
Caliber	84mm
Weight	16 kg with mount and telescopic sight
Length	1065 mm
Range	• HEAT - 400m (moving targets) and 500 m (stationery targets). • HE - 1000m (killing area 10 m). • Smoke - 1300 m (width 15 m). • Illumination - 2100 m (area of 400-500 m diameter for 30 seconds).
Rate of Fire	Maximum sustained 06 rounds per minute
Armor Penetration	400 mm (Heat)
Types of ammunition	Heat, HE, Smoke, Illumination
Back Blast Area	15 metres



COMPANY SUPPORT WEAPONS

30MM AUTOMATIC GRENADE LAUNCHER (AGL)

SPECIFICATION/ FEATURES

Manufacturer	Indian Ordnance Factory Board
Weight Launcher	14.2 kg
Weight Mount	10.2 kg
No. of rounds in belt	235 rounds
Range of Fire	800 to 1700 metre (with or without sight)
Rate of Fire	
Normal	50 Grenades/minute
Rapid	100 Grenades /minute
Cyclic	350 to 400 Grenades /minute



7.62MM MMG (MEDIUM MACHINE GUN)

SPECIFICATION/ FEATURES

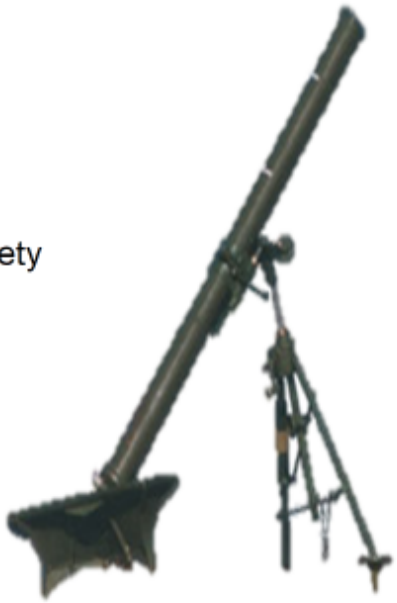
Manufacturer	Indian Ordnance Factory Board
Weight Gun	14.2 kg
Weight Tripod	10.2 kg
No. of rounds in belt	235 rounds
Effective range	1800 metre
Rate of Fire	
Normal	100 rounds/minute
Rapid	200 rounds/minute
Cyclic	500-1000 rounds/minute
Sustained Fire	Being air-cooled can maintain normal rate of fire indefinitely. However, barrel should be changed after firing four belts.



BATTALION SUPPORT WEAPONS
81MM MORTAR

SPECIFICATION/ FEATURES

Manufacturer	Indian Ordnance Factory Board
Caliber	81mm
Weight	40.6 kg (without sight). 4.7 kg (sight without case)
Muzzle velocity	305 metre/sec
Range	Minimum – 68 meters (from safety point of view 90 m). Maximum – 5200 metres
Rate of Fire	
Slow	6 – 8 rounds/minute
Normal	9– 11 rounds/minute
Rapid	12-20 rounds/minute
Safety Distance	Flanking 200 metre Overhead 250 metres



ANTI-TANK GUIDED MISSILE (MILAN-2T)

SPECIFICATION/ FEATURES

Milan-2T is a man portable second generation ATGM, to destroy tanks fitted with Explosive Reactive Armour, moving and stationary targets

Manufacturer	Bharat Dynamics Limited
Range	25 metre - 2000 metre
Weight of Launcher	Missile in Logistic Container 80 kg
	Missile in Tactical Packing-11.8 kg
Missile Length	769mm
Launch Tube Length	1200mm
Caliber	115mm
Operating Temp	40 to 50 degree C
Hit Probability	90 to 95% (more then 300m)
Velocity	75 – 210 m/sec
Time to Flight	12.5 sec



CONCLUSION

13. Infantry Battalion, the cadets would have realised is the bed rock of the Indian Army. Highly trained and motivated, the Infantry man stands guarding our borders 24x7 and 365 days in a year. Today, Infantry Battalion has state of art weapons and equipment details of which have been taught to the cadets. The modernization is a continuous process and, in the days, ahead it will only be better. However, it is not only the technology that matters. What matters most is the Infantryman who uses the weapon with lethality. The Infantry is the Cutting Edge of the Indian Army and is ready for future wars.

SUMMARY

- **5.56 INSAS Rifle**. A lightweight, compact, and low-recoil personal weapon used by the Indian Army, part of the Indian Small Arms System (INSAS). It is 25% lighter and has 70% less recoil than the 7.62 mm SLR, making it easier to handle and fire.

- **Ammunition Types**.
  - Ball rounds
  - Tracer rounds
  - Blank rounds
  - High-density cartridges

- **Infantry Battalion**. The basic fighting unit of the Indian Army, capable of operating independently or as part of a larger force. Its strength comprises 20 officers, 42 JCOs, and 794 other

ranks:-

- Smallest Unit: A section.
- Capabilities:
- Self-reliance
- Ability to hold ground
- Adaptability to various situations
- High mobility

➢ **Company Support Weapons**. Essential for enhancing combat effectiveness, including:-

- 7.62 mm Dragunov Sniper Rifle
- 7.62 mm Medium Machine Gun
- 30 mm Medium Grenade Launcher
- 84 mm Rocket Launcher

➢ **Battalion Support Weapons**. These weapons are critical in causing maximum casualties to the enemy and are extensively used in offensive, defensive, and special operations:

- Types.
- 81 mm Mortar
- Anti-Tank Guided Missile (ATGM)
- Key Features: Include weight, range, rate of fire, muzzle velocity, and elevation, essential for cadets to understand.

ASSESSMENT EXERCISE

Multiple Choice Question

Q1. What is the basic fighting unit of the Indian Army?

(a) Section
(b) Infantry Battalion
(c) Brigade
(d) Division

Q2. How many officers are typically present in an Infantry Battalion?

(a) 15
(b) 19
(c) 25
(d) 30

Q3. What is the smallest fighting unit of an Infantry Battalion?

(a) Platoon
(b) Section
(c) Company
(d) Regiment

Q4. Which of the following is a capability of an Infantry Battalion?

(a) Flying
(b) Self-reliance
(c) Heavy artillery operation
(d) None of the above

Q5. Which rifle is part of the Indian Small Arms System (INSAS)?

(a) 7.62 mm SLR
(b) 5.56 INSAS Rifle
(c) AK-47
(d) Dragunov Sniper Rifle

Q6. What type of ammunition is not used with the 5.56 INSAS Rifle?

(a) Ball round
(b) Blank round
(c) High-density cartridge
(d) Explosive round

Q7. What is the range of the 7.62 mm Dragunov Sniper Rifle?

(a) Short-range
(b) Medium-range
(c) Long-range
(d) Ultra-long-range

Q8. Which of these is a company support weapon?

(a) 81 mm Mortar
(b) Anti-Tank Guided Missile
(c) 30 mm Medium Grenade Launcher
(d) Swathi Radar System

Q9. What is the primary purpose of an 81 mm Mortar?

(a) Anti-aircraft defence
(b) Heavy artillery support
(c) Indirect fire support
(d) Close combat

Q10. Which weapon system is primarily used against tanks?

(a) Rocket Launcher
(b) 81 mm Mortar
(c) Anti-Tank Guided Missile
(d) Medium Machine Gun

Q11. What does INSAS stand for?

(a) Indian National Small Arms System
(b) Indian Small Arms System
(c) Infantry Small Arms System
(d) International Small Arms System

Q12. Which capability is NOT associated with an Infantry Battalion?

(a) Mobility
(b) Adaptability
(c) Maritime warfare
(d) Holding ground

Q13. What is the recoil percentage reduction of the 5.56 INSAS Rifle compared to the 7.62 mm SLR?

(a) 50%
(b) 60%
(c) 70%
(d) 80%

Q14. Which company support weapon is used for precision targeting?

(a) Medium Machine Gun
(b) Dragunov Sniper Rifle
(c) Grenade Launcher
(d) Rocket Launcher

Q15. What is the main feature evaluated in Infantry Battalion support weapons?

(a) Cost
(b) Elevation
(c) Weight and range
(d) Both (b) and (c)

Short Answer Type

Q1. What are the types of ammunition used with the 5.56 INSAS Rifle?

Q2. What are the key capabilities of an Infantry Battalion?

Long Answer Type

Q1. Explain the structure and capabilities of an Infantry Battalion in detail.

Q2. Describe the role and characteristics of company support weapons in an Infantry Battalion.

Q3. Discuss the types and features of Infantry Battalion support weapons.

SPECIALISED SUBJECT (ARMY)

MWE-II: MILITARY WEAPONS & EQUIPMENTS (SD/SW)

"Weapons are tools of peace when held with discipline, but tools of destruction when used without wisdom."

"वीरता धर्मश्च शस्त्रस्य भूषणम्।"

(Courage and righteousness are the ornaments of a weapon)



TEACHING INSTRUCTIONS

Period	**:**	**04 (160 Mins)**
Type	**:**	**Lecture**
Year	**:**	**1st Year (01) & 2nd Year (03)**
Conducting Officer	**:**	**Permanent Instructor**

Training Aids: Class Room, OHP, Board and Chalk/Markers

Time Plan

• **Introduction**	**:**	**02 Mins**
• **Part I**	**:**	**38 Mins**
• **Part II**	**:**	**40 Mins**
• **Part III**	**:**	**35 Mins**
• **Part IV**	**:**	**40 Mins**
• **Conclusion & Summary**	**:**	**05 Min**

INTRODUCTION

1. It is essential for any responsible citizen to understand the key weapon systems used by the Indian Army, which are integral to its operations and national defence. Therefore, the advanced technology and importance of these systems in safeguarding our country must be well understood by cadets. It is also imperative for the cadets to understand that in today's world being militarily strong is a necessity to protect the sovereignty of a country from external and internal threats. Towards that end, in the last decade or so India in its call for "ATMANIRBHAR BHARAT" has substantially increased indigenous production of military equipment. In this lesson, the cadets will explore operational capabilities of major weapon systems, such as tanks, artillery, missiles and aircraft, which form the backbone of India's military strength.

2. The lesson will give a brief of how these weapons are designed, their unique features, and the role they play in modern warfare. By understanding their functionality, the cadets will gain insight into how these systems support both offensive and defensive military operations. Additionally, their role in achieving military objectives from battlefield superiority to deterrence, and how they help maintain peace and security will be studied. The lesson aims to build technical knowledge and foster an appreciation for the dedication and discipline required to operate these systems effectively.

INTERESTING FACTS

- **The Brahmos missile took just 4 years (from 2001 to 2005) to develop, making it one of the fastest missile development programs in the world Brahmos missile is named after Brahmaputra River (India) and Moskva River (Russia), this missile became the first supersonic cruise missile in the world to be launched from a submarine in 2013**.



PREVIEW

The lecture will be conducted in the following parts:

- **Part I: Armoured & Mechanised Infantry.**
- **Part II: Artillery & Rockets.**
- **Part III: AAD & Missile System.**
- **Part IV: UAVs & Major Equipment.**

LEARNING OBJECTIVES

- **Be aware of major weapons & equipment being used by the Indian Army**.
- **Understands the operational capabilities of weapons & appreciate their role in defending the nation from external threats**.
- **The relevance of 'Make in India'**.

PART I: ARMOURED AND MECHANISED INFANTRY

3. **Tanks**. Tanks are heavily armoured combat vehicles with powerful guns, forming the backbone of mechanized Infantry and armoured divisions. Designed for direct fire combat, they excel in breaking enemy lines and manoeuvring over rough terrain.

4. **Key Features**:

(a) **Armour**. Thick steel/composite protection against anti-tank weapons.

(b) **Weaponry**. Large-calibre main gun (105mm–125mm) with secondary armament.

(c) **Mobility**. Tracks enable movement across difficult terrains.

(d) **Crew**. Typically consists of a driver, gunner, loader, and commander.

(e) **Role**. Provides offensive firepower, supports Infantry, and leads breakthroughs.

5. **BMP (Infantry Fighting Vehicle)**. The BMP is a Soviet-designed Infantry fighting vehicle that transports troops while providing fire support. Unlike APCs, it is armed to engage targets independently, balancing mobility and firepower.

6. **Key Features**:

(a) **Armour**. Lighter than tanks but protects against small arms and shrapnel.

(b) **Weaponry**. Equipped with automatic cannons, ATGMs, and machine guns.

(c) **Mobility**. Tracks ensure excellent cross-terrain movement.

(d) **Role**. Transports Infantry and offers fire support.

(e) **Crew & Infantry**. Holds 3-4 crew members and 6-8 Infantrymen.

FEATURES	SPECIFICATION
CREW	3 (COMMANDER AND TWO GUNNER)
WEIGHT	46 TONS
MAX SPEED	60 KM/HR
RANGE	550 KM
ARMAMENT	125mm gun (Main Weapon), 7.62mm Coaxial MG, 12.7mm MG,7.62 mm PKMT
AMMUNITION	HESH, HEAT, MISSILE, SMOKE, ILLUMINATION



7. **T 90 Main Battle Tank (Bhishma Indian Version)**. T-90 is a cutting-edge main battle tank, originally developed in Russia and later upgraded for the Indian Army as "Bhishma." It combines powerful firepower, mobility, and protection, making it a key asset in modern warfare. It excels in both offensive and defensive operations. Its adaptability to extreme weather and challenging terrains has been proven in combat zones like Rajasthan's deserts and Ladakh's high altitudes.

DID YOU KNOW?

- The T-90 Bhishma uses Kontakt-5 explosive reactive armour, which can neutralize the impact of enemy projectiles, including advanced kinetic energy penetrators.
- The T-90 Bhishma is equipped with the Shtora-1 system, a defensive technology designed to jam the guidance systems of incoming anti-tank missiles, significantly increasing its survivability in combat.
- T-90 Bhishma has been strategically deployed in high-altitude areas like Eastern Ladakh during tensions with neighbouring countries, proving its adaptability to extreme conditions.

8. **Main Battle Tank (MBT) ARJUN** India's indigenously developed third-generation main battle tank, the MBT Arjun, is designed by DRDO to enhance the Indian Army's armoured warfare capabilities. Named after the Mahabharata warrior Arjuna, it symbolizes strength, precision, and innovation. A key asset in the armoured corps, the Arjun excels in offensive and defensive operations. With advanced technology, superior firepower, and robust armour, it performs effectively in diverse environments, including Rajasthan's deserts and semi-urban terrains. Its precision targeting and strong protection make it invaluable in sustained combat scenarios.

FEATURE	SPECIFICATION
CREW	4
WEIGHT	58.5 Tons
MAX SPEED	60 Kmph
RANGE	450 km
ARMAMENT	120 mm Rifled Gun (Main Weapon), 7.62 mm Coaxial MG, 12.7 mm Anti-Aircraft MG
AMMUNITION	HESH, HEAT, Smoke, Illumination.



DID YOU KNOW?

- During comparative trials conducted in 2010, the Arjun outperformed the T-90 in firepower, accuracy, and mobility, proving its capability as a modern battle tank.
- The Arjun tank's commander's panoramic sight allows 360-degree surveillance, providing superior battlefield awareness for its crew.
- Arjun's hydropneumatics suspension is not only a rarity in tanks but also provides exceptional ride quality and stability, even on rough terrains.

9. **ICV BMP2 Infantry Fighting Vehicle (IFV)**. The Soviet-designed BMP-2 is a key IFV in the Indian Army, used by Mechanized Infantry Regiments. It transports Infantry while providing fire support and protection, upgraded to meet Indian Army requirements.

10. Key Features of IFV are as under:-

(a) **Role**. Infantry transport, fire support.

(b) **Armament**. 30mm automatic cannon, 7.62mm coaxial machine gun, AT-5 Spandrel missiles.

(c) **Armour**. Protects against small arms fire, shell splinters, and some larger projectiles.

(d) **Mobility**. Tracked, high road speed, amphibious capability.

FEATURE	SPECIFICATION
TYPE	Infantry Fighting Vehicle (IFV)
WEIGHT	14.3 tons
CREW	3 + 7 passengers
RANGE	600 km (road)
ANTI-TANK CAPABILITY	2 x AT-5 Spandrel (ATGM)
GUIDANCE SYSTEM FOR ATGM	Semi-automatic command-to-line-of-sight (SACLOS)
NBC PROTECTION	Yes
SERVICE ENTRY (INDIAN ARMY)	1980s



11. **Applications in Indian Army are as under:-**

(a) The BMP-2 is a crucial component for Indian Mechanized Infantry, providing quick mobility, firepower and protection.

(b) The Indian Army has upgraded BMP-2 with better armour, enhanced fire control systems and advanced communications for modern warfare.

DID YOU KNOW?

- The BMP-2 was extensively used during the Kargil War (1999) to provide fire support to Infantry units in high-altitude combat, proving its effectiveness even in extreme environments.
- The BMP-2 is designed to be air-transportable and can be quickly deployed using IL-76 or C-17 Globemaster aircraft, making it a rapid-response asset in emergencies.
- Future upgrades for the BMP-2 include remote weapon stations, where the gunner can operate the armament without exposing themselves to enemy fire.

PART II: ARTILLERY AND ROCKETS

12. **Artillery in the Indian Army**. Artillery is a key component of the Indian Army, providing long-range fire support to Infantry and armoured units. It includes large-calibre guns, howitzers, and mortars, capable of delivering powerful firepower at extended distances. Artillery is used for bombardment, counter-battery fire, and supporting offensive and defensive operations.

13. The Indian Army's Artillery is undergoing major upgradation. Some of the equipment being used at present are as under:-

(a) **155mm Howitzers**. M777 and Dhanush, for precision fire support.

(b) **Medium & Heavy Mortar 120mm**. For close support and high-angle fire.

(c) **Field Guns**. Used in direct-fire support.

Rockets in the Indian Army

14. The Indian Army also uses multiple rocket launcher systems (MRLS) for mass fire support. These systems are designed to deliver a high volume of rockets quickly over a large area, causing significant damage to enemy formations.

15. Prominent rocket systems used by Artillery are as under:-

(a) **Pinaka**. A multi-barrel rocket launcher system, capable of delivering devastating barrages at a range of 40-75 km.

(b) **Smerch**. A Russian-made long-range rocket launcher, effective at targeting large enemy formations.

16. These artillery and rocket systems play a crucial role in providing firepower, enhancing the Indian Army's operational capabilities.



17. **155 mm Artillery Gun (BOFORS)**. The 155mm Bofors Gun is a highly effective field artillery weapon used by the Indian Army. Known for its exceptional accuracy, range, and firepower, it has been a game-changer in modern artillery warfare. The gun is designed to provide long-range fire support and has proven its capabilities in both defensive and offensive operations.

18. The 155mm Bofors Gun is a cornerstone of the Indian Artillery's firepower, renowned for its reliability and effectiveness in both conventional and counter-insurgency operations. Its ability to deliver sustained and accurate firepower makes it indispensable in achieving battlefield dominance.

FEATURES	SPECIFICATIONS
TYPE	Long range fire support, precision bombardment, counter-battery fire
SHELL TYPES	High explosive, smoke, and illumination rounds
CALIBER	155mm
MOBILITY	Towered artillery system with excellent maneuverability across varied terrains
RANGE	Effective range: Up to 30km Extended range: Up to 40km with rocket assisted projectile
CREW	6-8 personnel
SPECIAL FEATURE	Semi-automatic loading system for higher rates of fire
RATE OF FIRE	3 rounds in 15 seconds



DID YOU KNOW?

- The 155mm Bofors gun played a crucial role in the 1999 Kargil War, effectively targeting enemy positions in high-altitude terrain.
- The gun is designed for high rates of fire, enabling it to maintain prolonged bombardment during intense battlefield scenarios.

19. **Pinaka Multi Barrel Rocket Launcher**. The Pinaka Multi-Barrel Rocket Launcher (MBRL) is an indigenous artillery system developed by DRDO for the Indian Army. Named after Lord Shiva's bow, it delivers heavy firepower rapidly, targeting critical enemy positions. Used effectively in the 1999 Kargil War, it fires guided or unguided rockets. The advanced Mark-II version offers improved range and accuracy, enhancing its strategic value.

(a) **High Firepower**. Pinaka can fire a volley of 12 rockets within 44 seconds, effectively delivering a large amount of firepower in a short time.

(b) **Extended Range**. The Mark-II version has an extended range of up to 75 km, giving it a significant advantage in modern combat situations.

(c) **Multiple Warheads**. Capable of firing different types of warheads, including fragmentation and high-explosive, making it versatile for various combat scenarios.

(d) **Mobile and Rapid Deployment**. The system is mounted on a mobile platform, enabling rapid deployment in battlefield condition.

20. PINAKA Rocket system is a state of art equipment. Applications of PINAKA Rocket System in the Indian Army are as under:-

(a) **Long-Range Fire Support**. PINAKA provides critical long-range firepower to neutralize enemy positions, troop concentrations, and supply lines up to 75 km.

(b) **Counter-Battery Operations**. It is highly effective in targeting enemy artillery and rocket launchers, ensuring swift suppression of hostile fire.

(c) **Precision Strikes**. The guided version enables precise targeting of high-value enemy assets like bunkers, radar stations, and command posts.

(d) **High-Altitude and Extreme Terrain Warfare**. PINAKA's adaptability makes it suitable for operations in challenging terrains, including Ladakh and the deserts of Rajasthan.

(e) **Mobility and Shoot-and-Scoot Capability**. Mounted on a mobile platform, it supports rapid deployment and quick relocation to evade enemy counterfire.

(f) **Force Multiplier**. With diverse warheads, including high-explosive, cluster, and anti-tank munitions, it significantly enhances operational flexibility.

(g) **Suppression of Enemy Air Defence (SEAD)**. PINAKA aids in clearing enemy air defence systems, enabling safe aerial operations.



FEATURES	SPECIFICATIONS
RANGE	MarkI-40km Mark II-60km
WEIGHT	15 tons (launcher)
AMMUNITION TYPE	214 mm Unguided Rkts.
RATE OF FIRE	Fires 12 Rkts in 44 Sec

DID YOU KNOW?

- The system is named after the bow of Lord Shiva, symbolizing its immense power and precision.
- Its ability to target multiple locations simultaneously, PINAKA can suppress large enemy formations and infrastructure in a single barrage.
- Despite being a land-based system, PINAKA can operate in amphibious warfare scenarios, thanks to its mobile launch platform.

PART III: AAD SYSTEM & MISSILE

21. **Missile Systems in the Indian Armed Forces**. The Indian Armed Forces employ a wide range of advanced missile systems for strategic defence and combat operations. These missile systems are designed to target land, air, and sea threats with precision and speed, enhancing India's defence capabilities.

22. **Key missile systems include**:

(a) **Surface-to-Air Missiles (SAMs)**. Like the Akash and Barak-8, designed to protect airspace by targeting enemy aircraft and missiles.

(b) **Surface-to-Surface Missiles (SSMs)**. Such as the Prithvi and Agni series, which provide strategic and tactical strike capabilities against ground targets.

(c) **Anti-Tank Guided Missiles (ATGMs)**. Like the Nag missile, used to target and destroy enemy tanks and armoured vehicles.

(d) **Cruise Missiles**. The BrahMos, a supersonic cruise missile, capable of striking both land and sea targets with high precision.

FEATURE	AGNI I	AGNI II	AGNI III	AGNI IV	AGNI V
RANGE	700KM	2,000KM	3,500KM	4,000KM	5,000+KM
PAYLOAD	1,000kg (Nuclear or conventional)	1,000-1,500 kg	1,500kg	1,500kg	1,500kg
WARHAD	Nuclear Conventional	Nuclear Conventional	Nuclear Conventional	Nuclear Conventional	Nuclear Conventional
PROPELLANT	Solid Propellant	Solid Propellant	Solid Propellant	Solid Propellant	Solid Propellant
LAUNCH PLATFORM	Mobile/Truck based	Mobile/Truck based	Mobile/Truck based	Mobile/Truck based	Mobile/Truck based
GUIDANCE SYSTEM	Internal Navigation GPS	Internal Navigation GPS	Internal Navigation GPS	Internal Navigation GPS	Internal Navigation GPS
SPEED	Mach 5	Mach 5	Mach 7	Mach 7	Mach 24
INDUCTION YEAR	2000	2002	2007	2012	2015

These missile systems are integral to the Indian Armed Forces' deterrence strategy, ensuring national security and operational effectiveness.

Surface-to-Surface Missile System (SSMs)

23. **Agni Surface-to-Surface Missile System**. The Agni Missile System is a series of long-range ballistic missiles developed for India's defence. Capable of carrying conventional and nuclear warheads, it ranges from a few hundred to over 5,000 km, ensuring strategic deterrence.

24. **Prithvi Surface - to - Surface Missile System**. The Prithvi Missile System is India's first indigenously developed ballistic missile series, designed for tactical battlefield operations. It is part of the Integrated Guided Missile Development Program (IGMDP) and serves as a versatile weapon for delivering precise strikes with both conventional and nuclear warheads.



FEATURE	PRITHVI I	PRITHVI II	PRITHVI III
RANGE	150 km	250 km	350km
PAYLOAD	1,000kg (Nuclear Conventional)	500-1000kg (Nuclear Convention)	500-1000kg (Nuclear Convention)
WARHEAD	Nuclear Conventional	Nuclear Conventional	Nuclear Conventional
PROPELLANT	Liquid Propellant	Liquid Propellant	Liquid Propellant
LAUNCH PLATFORM	Mobile/Truck based	Mobile/Truck based	Mobile/truck based
GUIDANCE SYSTEM	Inertial Navigation	Inertial Navigation GPS	Inertial Navigation GPS
SPEED	Mach 5	Mach 5	Mach 5
INDUCTION YEAR	1988	2003	2004

25. **BrahMos Missile System**. The BrahMos Missile is a supersonic cruise missile jointly developed by India and Russia. It is recognized for its high speed, precision.

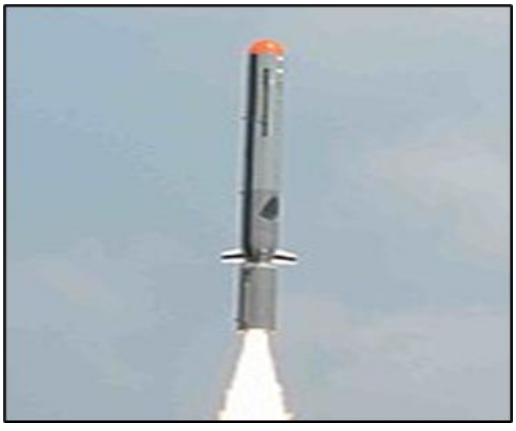




FEATURES	SPECIFICATION
TYPE	Supersonic cruise missile
RANGE	VERSION I RANGE: 290km VERSION II RANGE: 450km (Can vary depending on launch platform)
WEIGHT	3,000kg (Including Warhead)
AMMUNITION TYPE	Conventional warheads upto 300 kg (High explosive fragmentation)
RATE OF FIRE	N/A (Single shot, Cruise missile)
SPECIAL QUALITIES/ FEATURES	- SPEED: Mach 2.8 to Mach 3 - Highly accurate with terminal guidance - Versatile launch - Platforms (Air, Land, Sea and Submarines) - Advanced stealth technology - Quick reaction time and reduced radar detection

26. **Nirbhay Missile System**. The Nirbhay Missile is a long-range, subsonic cruise missile developed by India to provide precision strike capability in tactical and strategic scenarios. It is designed to fly at low altitudes, avoiding radar detection and enhancing survivability. It adds flexibility to India's defence strategy, complementing other missile systems with its long-range and stealth capabilities.

FEATURES	SPECIFICATIONS
Range	Up to 1,500 km, suitable for deep penetration
Capabilities	Terrain-hugging flight for stealth operations
Payload	Can carry both conventional and nuclear warheads



27. **Anti-Tank Guided Missiles (ATGMs) NAG Missile System**. The Nag Missile is an indigenously developed anti-tank guided missile (ATGM) designed to neutralize heavily armoured vehicles with precision. It is a fire-and-forget missile, capable of operating in diverse battlefield conditions.

FEATURES	SPECIFICATIONS
Range	Up to 4 km(land-based), and 7 km(air borne)
Guidance System	Advanced imaging infrared seeker for accuracy
Launch Platforms	Deployed from specially designed vehicles or helicopters
Other Features	Fire & Forget technologies, high penetration capabilities, versality, day/night capability



28. **L-70 Anti-Aircraft Gun**. The **L-70 Anti-Aircraft Gun** is a reliable, towed anti-aircraft weapon used extensively by India's Air Defence Regiments. Originally developed by Bofors, it has been modernized to meet contemporary battlefield requirements, making it effective against low-flying aerial threats like drones, helicopters, and fighter jets.

FEATURES	SPECIFICATIONS
RATE OF FIRE	Up to 300 rounds per minute (ground targets) Up to 240 rounds per minute (aerial targets)
RANGE	Up to 4 km for aerial targets
MOBILITY	Can be deployed in various locations including border and key installations
LOW ALTITUDE AIR DEFENCE	Ideal for engaging enemy helicopters and drones
CALIBER	40 mm



PART IV: UAVs & MAJOR EQUIPMENT IN INDIAN ARMY

29. **UAVs in Indian Defence Forces**. Unmanned Aerial Vehicles (UAVs), commonly known as drones, have become integral to modern warfare. The Indian Defence Forces utilize various UAVs for reconnaissance, surveillance, target acquisition, and strike capabilities, enhancing operational effectiveness in diverse terrains and conditions. They provide the Indian Defence Forces with unmatched flexibility for surveillance, intelligence

gathering, and precision strikes, playing a pivotal role in modernizing India's military operations.





KEY UAVs IN INDIAN DEFENCE FORCES

	SEARCHER MKII	HERON	RUSTOM II	HAROP
ROLE	Reconnaissance and surveillance	Medium-altitude long-endurance(MALE) UAV	Indigenious MALE UAV for surveillance and armed roles	Loitering munition for precision strikes
RANGE	Upto 300 km	Up to 1000km	250 km	1000+ km
ENDURANCE	16-18 hours	Over 30 hrs	24 hrs	6 hours in loitering mode
PAYLOAD	Electro-optical and infrared sensors	Electro-optical, infrared, and synthetic aperture radar	Advanced imaging systems	16kg of warhead
CAPABILITIES	Terrain-following and target tracking, suitable for high altitude operations	Real-time intelligence and target acquisition	Autonomous takeoff and landing capabilities	Directly strikes enemy targets like radars and missile sites

30. **Swathi Weapon Locating Radar (WLR)**. The Swathi Weapon Locating Radar (WLR), developed by DRDO, provides real-time enemy artillery tracking. This counter battery is designed to detect and track incoming artillery and rocket fire to determine the point of origin for counter battle fire. Manufactured by Bharat Electronics Limited,it is a key asset and showcases India's defence innovation.



FEATURES	SPECIFICATIONS
ACCURATE TARGETING	Swathi pinpoints enemy artillery, mortars and rockets for precise counter-attacks.
MULTI TARGET TRACKING	Tracks multiple projectiles in high-intensity conflicts
HIGH MOBILITY	Mounted on a rugged vehicle for rapid deployment across diverse terrains
ALL-WEATHER OPERATIONS	Functions effectively in extreme conditions, ensuring 24/7 readiness
QUICK SETUP	Becomes operational within 30 mins for rapid response

31. **Bridge Layer Tank**. The **Bridge Layer Tank (BLT)** is an essential engineering vehicle used by the Indian Army to facilitate rapid mobility across obstacles such as rivers, ditches, and other water bodies during combat operations. It is equipped with a fully operational bridge that can be deployed and retracted quickly, allowing armoured and mechanized units to cross difficult terrain with minimal delay. The BLT ensures seamless manoeuvrability of forces, providing strategic advantage during offensive and defensive operations.



FEATURES	SPECIFICATIONS
RAPID DEPLOYMENT	Deploys bridges within minutes for troop and vehicle
HEAVY LOAD	Supports tanks and heavy mechanized units
HIGH MOBILITY	Mounted on a tank chassis & operates in rough terrains
CREW PROTECTION	Armoured chasis resists small arms fire and shell fragments
INTEGRATION WITH ARMOURED UNIT	Ensures smooth movement
ALL-TERRAIN CAPABILITIES	Functions effectively in mud, sand and snow

32. **PMP&PMS Bridge System**. The **PMP and PMS Bridge Systems** are advanced modular bridges enabling quick troop and vehicle movement across water obstacles and tough terrains. Designed for rapid deployment, they enhance operational mobility and sustain combat momentum.



PARAMETERS	DETAILS
FLEXIBLE DESIGN	Modular components allow varied bridge lengths
HIGH LOAD CAPACITY	Supports tanks and mechanized units.
RAPID DEPLOYMENT	Quick assembly and dissemble for swift crossings
TERRAIN ADAPTABILITY	Functions on rivers, canals and uneven ground
EASY TRANSPORT	Moves easily via trucks and military vehicles
DURABLE BUILD	Corrosion resistant for long term use
COMBAT INTEGRATION	Supports tactical maneuver and unit mobility

33. **Differences between PMP and PMS**. The major differences between PMP and PMS is shown in tabulated form below.

	PMP BRIDGE SYSTEM	PMS BRIDGE SYSTEM
BRIDGE LENGTH	Up to 60 meters	Up to 100 meters
LOAD CAPACITY	Supports up to 60 tons	Supports mechanized Infantry and tanks
DEPLOYMENT TIME	Approximately 2-3 hours	2 hours under ideal conditions

34. **Advanced Light Helicopter (ALH) DHRUV**. The **Advanced Light Helicopter (ALH) Dhruv** is an indigenously designed and manufactured multi-role helicopter developed by Hindustan Aeronautics Limited (HAL). It serves as a versatile platform in the Indian Armed Forces, supporting a wide range of operations, including transport, reconnaissance, search and rescue, casualty evacuation, and combat missions. Known for its adaptability, Dhruv has become a critical asset across all branches of India's defence forces.



PARAMETERS	DETAILS
VERSATILE ROLES	Used for troop transport, logistics, medical evacuation and anti-insurgency
HIGH CAPACITY	Carries upto 4 troops or equivalent cargo
CUSTOMIZABLE	Available in military, naval and coast guard versions, armed or unarmed
RESCUE OPERATIONS	Deployment for disaster relief, search and aid transport
COMBAT-READY	Armed variants feature machine guns, rocket pods and anti-tank missiles
CRUISING SPEED	250 km/hr
RANGE	Up to 630km
PAYLOAD	1,500kg of cargo or 14 personnel

CONCLUSION

35. India has the fourth most powerful military in the world with strong Army, Navy and Air Force, India's defence budget for Financial Year 2025-26 was approx. 79 billion dollars. It is imperative that India maintained a strong Defence Force as India has volatile neighbouring countries. To keep them at bay it is imperative that our country maintains state of art weapons and equipment.

36. With government focussing on "Aatm Nirbhar Bharat", indigenous defence production has substantially grown. Project like MBT Arjun and Tejas Mk I have been a huge success wherein not only have been inducted in Armed Forces but also have been exported to foreign nations.

SUMMARY

- **Tanks & Infantry Fighting Vehicles (IFVs)**
  - T-90 Bhishma: Russian-origin MBT with a 125mm gun, advanced sensors, and ERA armour, effective in varied terrains.
  - MBT Arjun: Indigenous third-gen MBT with a 120mm rifled gun, compos ite armour, and advanced fire control.
  - BMP-2 IFV: Amphibious Soviet-designed vehicle with a 30mm cannon, anti-tank missiles, and machine guns.
- **Artillery & Rocket Systems**
  - 150mm BOFORS Gun: Long-range fire support with high-explosive and armour-piercing capabilities.
  - Pinaka MBRL: Indigenous rocket launcher with a 75 km range, used for saturation and precision strikes.
- **Missile Systems**
  - Agni Series: Long-range ballistic missiles for nuclear and conventional deterrence.
  - Prithvi Series: Short-range ballistic missiles for tactical operations with high precision.
  - BrahMos Cruise Missile: Supersonic, multi-platform missile with a 600 km range.
  - Nag ATGM: Fire-and-forget anti-tank missile for land and air platforms.
- **Air Defence & Surveillance**
  - L-70 Anti-Aircraft Gun: Upgraded system with radar and electro-optical targeting.
  - Swathi Radar: Tracks enemy artillery and rockets up to 50 km for counter-battery

fire.

- **Unmanned Aerial Vehicles (UAVs)**
  - Heron: Long-endurance surveillance UAV with a 30-hour flight time.
  - Searcher Mk II: Tactical UAV for intelligence gathering with a 300 km range.
  - Harop: Loitering munition for precision strikes on radar and missile systems.
- **Engineering & Mobility Systems**
  - BLT: Deploys 60-ton capacity bridges for mechanized units.
  - PMP & PMS Bridges: Modular systems for crossing difficult terrains.
- **Advanced Light Helicopter (ALH) Dhruv**
  - Multi-role HAL-developed helicopter for transport, evacuation, and combat.
  - Operates in high-altitude and adverse conditions with a 1,500 kg payload.

ASSESSMENT EXERCISE

Multiple Choice Questions (MCQs)

Q1. What is the primary role of the BMP-2 Infantry Fighting Vehicle?

(a) Air defence
(b) Infantry transport and fire support
(c) Long-range artillery
(d) Missile launching

Q2. Which system uses Kanchan composite armour?

(a) T-90 Bhishma
(b) MBT Arjun
(c) BMP-2
(d) Pinaka

Q3. What is the maximum range of the Pinaka Mark-II system?

(a) 40 km
(b) 60 km
(c) 75 km
(d) 90 km

Q4. What type of missile is the BrahMos?

(a) Ballistic
(b) Cruise
(c) Surface-to-air
(d) Anti-tank

Q5 What is the main armament of the T-90 Bhishma tank?

(a) 105mm rifled gun
(b) 120mm smoothbore gun
(c) 125mm smoothbore gun
(d) 130mm cannon

Q6. Which missile system is used for nuclear deterrence?

(a) BrahMos (b) Pinaka
(c) Agni (d) Nag

Q7. What is the operational range of the Swathi Weapon Locating Radar for detecting artillery?

(a) 30 km
(b) 40 km
(c) 50 km
(d) 60 km

Q8. Which UAV is used for loitering munitions in the Indian Army?

(a) Heron
(b) Rustom-II
(c) Harop
(d) Searcher Mk II

Q9. What is the deployment time for the Bridge Layer Tank (BLT)?

(a) 2–3 hours
(b) 10–15 minutes
(c) 5–10 minutes
(d) 20–30 minutes

Q10. Which system is equipped with the Shtora-1 defensive technology?

(a) MBT Arjun
(b) T-90 Bhishma
(c) BMP-2
(d) Pinaka

Q11. Which system in the Indian Army can fire 12 rockets in 44 seconds?

(a) Agni
(b) Pinaka
(c) Prithvi
(d) BrahMos

Q12. What is the main role of the Bridge Layer Tank (BLT)?

(a) Air defence
(b) Transporting Infantry
(c) Facilitating obstacle crossing
(d) Launching missiles

Q13. The Nag missile is designed for which specific purpose?

(a) Anti-aircraft operations
(b) Anti-tank operations
(c) Reconnaissance
(d) Strategic deterrence

Q14. What is the primary role of the Swathi Weapon Locating Radar?

(a) Detecting low-flying aircraft
(b) Locating enemy artillery
(c) Guiding UAVs
(d) Tracking ballistic missiles

Q15. Which missile system is capable of loitering before striking its target?

(a) Nirbhay
(b) Harop
(c) BrahMos
(d) Nag

Q16. What is the maximum operational altitude of the Advanced Light Helicopter Dhruv?

(a) 15,000 feet
(b) 18,000 feet
(c) 20,000 feet
(d) 25,000 feet

Q17. Which anti-aircraft gun system is extensively used by India's Air Defence Regiments?

(a) L-70
(b) ZU-23
(c) Akash SAM
(d) Barak-8

Q18. What is the key advantage of the BMP-2 over traditional APCs (Armoured Personnel Carriers)?

(a) Higher troop capacity
(b) Heavy artillery capabilities
(c) Armament for independent fire support
(d) Greater speed in water

Q19. Which UAV is primarily used for medium-altitude, long-endurance missions in the Indian Army?

(a) Rustom-II
(b) Searcher Mk II
(c) Heron
(d) Harop

Q20. What is the propulsion system of the BrahMos missile?

(a) Solid fuel
(b) Liquid fuel
(c) Solid and liquid fuel propulsion
(d) Ramjet engine

Short Answer Questions

Q1. What is the primary use of the Pinaka system?

Q2. Name two features of the MBT Arjun.

Q3. What role does the Swathi Weapon Locating Radar play?

Q4. What is the range of the BrahMos missile?

Q5. What is the main role of the BMP-2 in the Indian Army?

Long Answer Questions

Q1. Discuss the operational role and features of the MBT Arjun in the Indian Army.

Q2. Explain the significance of the Pinaka MBRL system in modern warfare.

Q3. Describe the role of UAVs in the Indian Army with examples.

Q4. What are the features and operational importance of the Swathi Weapon Locating Radar?

Q5. Compare the roles of the Agni and Prithvi missile systems in India's defence strategy

MWE –III: MODERN TRENDS IN WPN SYSTEMS & AMMUNITION (SD SW)

"BrahMos and Sudarshan are not just Weapons; They are Symbols of India's Journey From Ancient Strategy to Modern Supremacy"

विजयः संयुगे सत्यं, बलं विज्ञानसंयुतम्। *(Valmiki Ramayana, Aranya Kanda)*

"Victory in battle is assured when strength is united with knowledge."



TEACHING INSTRUCTIONS

Period	**:**	**02 (80 Mins)**
Type	**:**	**Lecture**
Year	**:**	**Third Year SD/SW**
Conducting Officer	**:**	**Permanent Instructor**

Training Aids: Class Room, OHP, Board and Chalk/ Markers

Time Plan

- **Introduction : 03 Mins**
- **Part - I : 37 Mins**
- **Part - II : 35 Mins**
- **Conclusion : 05 Mins**

INTRODUCTION

1. Imagine a battlefield where lasers disable enemy drones, robots clear landmines, soldiers use augmented reality headsets to navigate terrains, and drones silently hover over targets before striking with precision. This is not science fiction, it is the reality of modern warfare, shaped by groundbreaking advancements in military technology. For NCC cadets, understanding these technologies opens a window to the future of defence, showcasing how nations prepare to protect their sovereignty in an increasingly complex world.

2. The Indian Armed Forces, with a commitment to self-reliance, are rapidly adopting and developing such advanced systems. From directed energy weapons to AI-based situational awareness tools, these innovations enhance combat efficiency, reduce risks, and ensure informed decision-making. This session introduces these transformative technologies, highlighting their applications, advantages, and India-specific initiatives, giving cadets an exciting and insightful glimpse into the future of military operations.

INTERESTING FACTS

- **Laser Shields and Drone Hunters**: Did you know that lasers can act as shields? Directed energy weapons, like those being developed by the Indian Armed Forces, can instantly destroy enemy drones or missiles mid-air, protecting sensitive installations.
- **Robots in Combat**: Imagine robots clearing minefields or carrying supplies in rough terrains. The Indian Army already uses robots like Daksh for bomb disposal, paving the way for autonomous combat units in the future.
- **Augmented Reality on the Battlefield**: Soldiers are now equipped with wearable headsets that provide real-time 3D maps and enemy positions, almost like a video game but in real life. These technologies are being tested for use in Indian defence operations.



LEARNING OBJECTIVES

- Familiarize cadets with advanced military technologies and their applications.
- Develop an understanding of India-specific future inductions in defence technology.
- Inspire awareness about emerging trends in modern warfare.

- Part I: Advanced Military Technologies
- Part II: Emerging Technologies & Future Inductions

PART-I: MODERN TRENDS IN WPN SYSTEMS & AMN

3. **Directed Energy Weapons**. Directed Energy Weapons (DEWs) use focused energy beams such as lasers or microwaves to disable targets with high precision. These weapons have the advantage of silent and invisible operation, reducing collateral damage

and the cost per engagement. The Defence Research and Development Organisation (DRDO) in India has been working on DEWs, including the "Durga II" laser weapon system, intended to counter Unmanned Aerial Vehicles (UAVs) and other airborne threats.





SPECIFICATIONS	
MODEL OF DEVELOPMENT	Durga II (laser-based DEW)
EFFECTIVE RANGE	Typically, 1-2 km for current DEW prototypes
ENERGY SOURCE	High-power electricity or advanced batteries
TARGETING SYSTEM	Line-of-sight precision, allowing accurate engagement of moving targets
DEPLOYMENT PLATFORM	Intended for ground and vehicle-based platforms

Features and Capabilities

- Silent Engagement: Ideal for covert operations, minimizing detection risks.
- Minimal Collateral Damage: Targets specific threats without affecting nearby areas.
- Cost Efficiency: Minimal operational costs per use once deployed.
- Adaptable Use Cases: Effective against drones, UAVs, and electronic systems.

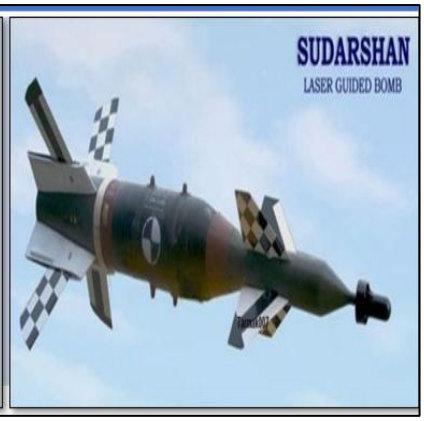
Applications

- Countering drones and missiles by neutralizing them mid-flight.
- Disabling electronic systems and surveillance equipment.
- Protecting sensitive installations from aerial threats.

4. **Precision Guided Munitions**. Precision Guided Munitions (PGMs) enable the Indian Armed Forces to conduct targeted strikes with minimal collateral damage. Equipped with advanced guidance systems like GPS and LASER targeting, PGMs ensure that strikes are accurate and lethal. The Indian Air Force and Army use systems like the Spice 2000 and Sudharshan Laser-Guided Bomb, providing accurate air-to-ground strike capabilities.







SPECIFICATIONS	
TYPES IN SERVICE	Spice 2000, Sudharshan Laser-Guided Bomb
GUIDANCE SYSTEM	GPS, laser, and infrared systems for precision
ACCURACY	Within a 1-meter radius
RANGE	Between 10 to 50 km, depending on the system
WARHEAD TYPES	Includes high-explosive, cluster munitions, and bunker buster options

Features and Capabilities

- High Precision Targeting: Capable of hitting exact locations with minimal error.
- Versatile Munition Types: Adaptable for different mission needs from anti-vehicle to anti-infrastructure.
- Day and Night Operations: Equipped to function effectively in all light conditions.
- Operational Versatility: Used for various tactical scenarios, including counter-terrorism and anti-infrastructure missions

India's Use

- Sudarshan laser-guided bombs for precision airstrikes.
- HELINA (Helicopter-Launched Nag Missile) for anti-tank operations.
- Smart Anti-Airfield Weapon (SAAW) for precision targeting of enemy runways.

5. **Loitering Munitions**. Loitering munitions, such as the Harop drones acquired by the Indian Armed Forces, combine drone and missile technologies to deliver precision strikes on high-value targets. These munitions can hover over a target area, waiting for the ideal opportunity to strike. Loitering munitions are particularly effective in time-sensitive missions where instant engagement is critical.

SPECIFICATIONS	
TYPE IN SERVICE	Harop Loitering Munition
ENDURANCE	Capable of loitering for up to 6 hours
RANGE	Effective up to 40 kilometers
WARHEAD TYPE	High-explosive payload
TARGETING SYSTEM	Electro-optical and infrared sensors

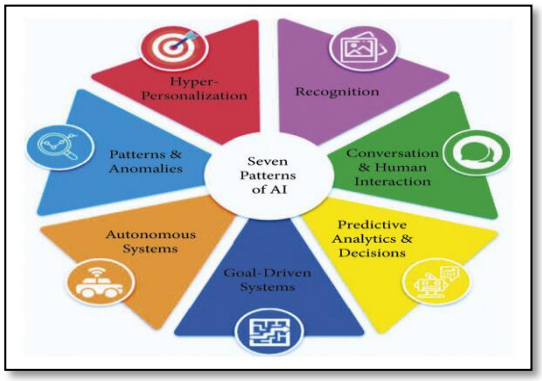
Features and Capabilities

- Real-Time Surveillance and Attack: Provides both intelligence gathering and strike capabilities.
- Instant Target Engagement: Suitable for quick-response scenarios.
- Autonomous Operation: Capable of engaging without direct operator control once deployed.
- Versatility in Use: Effective against enemy command posts, armoured vehicles, and radar installations.
- The Indian Armed Forces currently use loitering munitions such as "Nagastra 1", Warmat: and "ALS-50", which have ranges of up to 50 km, to target tanks, armoured vehicles and other assets.





6. **AI-Based Situational Awareness Systems**. AI-Based Situational Awareness Systems enhance battlefield decision-making by processing real-time data from various sources. These systems help Indian Armed Forces in reconnaissance, surveillance and target acquisition (RSTA) missions, enabling them to respond effectively to dynamic threats. The Combat Information Decision Support System (CIDSS) developed by DRDO exemplifies the use of AI in situational awareness, improving intelligence analysis and threat detection.



SPECIFICATIONS	
SYSTEM IN DEVELOPMENT	CIDSS
DATA INTEGRATION	Combines satellite, drone, and ground data
AI ALGORITHMS	Utilizes deep learning and computer vision for rapid data processing
RANGE	Limited only by the network of sensors deployed
PLATFORM	Usable on ground vehicles, drones, and command centres

Features and Capabilities

- Enhanced Threat Detection: Quickly identifies and categorizes threats, reducing response time.
- Improved Tactical Decisions: Assists commanders with detailed intelligence, improving mission outcomes.
- Adapts to Battle Changes: Responds to dynamic battlefield conditions in real-time.
- Multi-Source Data Analysis: Integrates multiple data feeds for a comprehensive battlefield view.

Applications

- Identifying threats and predicting enemy movements
- Providing actionable insights for informed decision-making

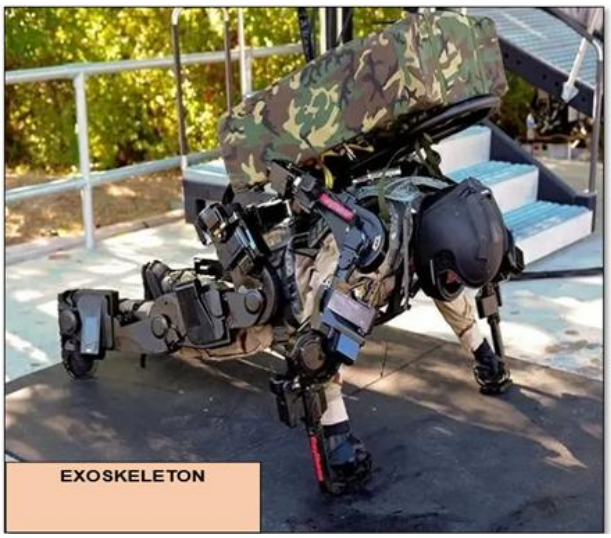
Advantages

- Reduces casualties and ensures equipment safety
- Enables 24/7 monitoring of conflict zones

India's Use

- DRDO's Combat Information Decision Support System (CIDSS) integrates AI for enhanced battlefield awareness.
- AI-enabled drones and surveillance systems for real-time data processing.

7. **<u>Military Robotics and Autonomous Systems</u>**. Military Robotics and Autonomous Systems are a growing area of focus in India's defence landscape. With developments like the DRDO "Daksh", a robot designed for Explosive Ordnance Disposal (EOD), these systems assist in dangerous missions, minimizing life threatening risk to personnel. Robotics enhance operational capabilities in the in the fields of reconnaissance and explosive disposal.





SPECIFICATIONS	
SYSTEM IN SERVICE	Daksh EOD Robot
CAPABILITIES	Can detect and defuse IEDs and landmines
MOBILITY	Operates on rugged terrain, suitable for combat zones
CONTROL RANGE	Operable remotely within a range of 500 meters
PAYLOAD CAPACITY	Equipped with tools for EOD operations

<u>Features and Capabilities</u>

- Reduces Soldier Casualties: Takes on high-risk tasks to protect personnel.
- Versatile Mission Support: From reconnaissance to bomb disposal.
- Remote Control Capability: Allows operators to manage operations safely from a distance.
- Adaptable in Varied Terrains: Designed to operate in rugged and hazardous environments.

Advantages

- Minimizes human exposure to dangerous environments.
- Increases operational efficiency in challenging terrains.

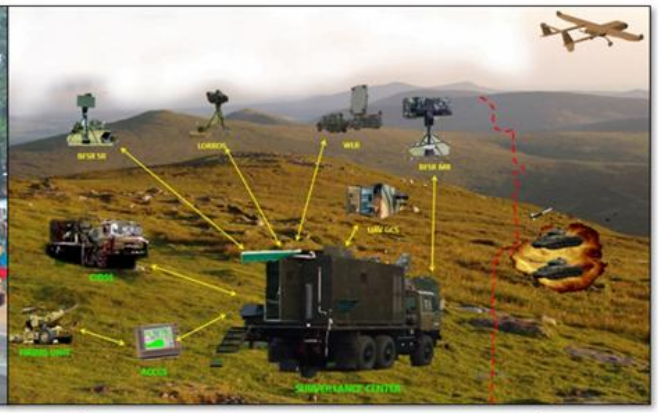
India's Use

- Daksh robot developed by DRDO for bomb disposal.
- Exoskeleton prototypes being tested for enhancing soldier endurance.

8. **INTERNET OF MILITARY THINGS (IoMT)**. The Internet of Military Things (IoMT) enables a cohesive, interconnected network of assets, providing a real-time operational picture. Through IoMT, the Indian Armed Forces aim to enhance battlefield awareness and decision-making. DRDO is working on integrating IoMT with systems like UAVs, tanks, and command centers, improving coordination and efficiency.



TACTICAL COMMUNICATION SSTEM



BATTLE FIELD SURVILLANCE SYSTEM

SPECIFICATIONS	
OPERATIONAL SCOPE	Network of sensors, communication devices and control systems.
DATA TRANSMISSION	Real-time data sharing across units.
COVERAGE RANGE	Variable, extending across battle zones through secure networks.
PLATFORM COMPATIBILITY	Operable remotely within a range of 500 meters.
PAYLOAD CAPACITY	Usable on drones, ground vehicles, ships, and command centres.
Features and Capabilities	
➢ Real-Time Connectivity: Enables synchronized operations across all units. ➢ Improved Situational Awareness: Provides up-to-date battlefield information. ➢ Enhanced Decision Support: Facilitates quick response to threats. ➢ Cyber-Resilient: Designed to withstand cyber threats and maintain data integrity.	
Advantages	
➢ Improves decision-making with a unified command network. ➢ Reduces response time during critical missions.	
India's Use	
➢ Tactical Communication Systems (TCS) and Battlefield Management Systems (BMS) focus on IoMT integration for real-time command and control	

9. **Cyber Warfare**. Cyber warfare capabilities are critical in contemporary military operations. They encompass offensive and defensive measures to attack and protect digital infrastructure. Offensive capabilities, such as malware and ransomware, can disrupt enemy

networks, while defensive strategies, like firewalls and intrusion detection systems, shield sensitive information and maintain operational integrity.



SPECIFICATIONS
➢ Offensive Tools: Malware, ransomware, phishing attacks, denial-of-service (DoS) attacks. ➢ Defensive Tools: Firewalls, intrusion detection systems (IDS), data encryption. ➢ Integration: Coordinated with intelligence and cyber command units. ➢ Capabilities: Offensive and defensive measures for network integrity, data security, and operational safety. ➢ Threat Detection: Advanced analytics and AI-powered threat detection. ➢ Capability to infiltrate enemy networks and disable critical infrastructure. ➢ Defensive systems to protect military databases and communications. ➢ Advanced encryption for securing data transmission. ➢ AI-driven threat detection and response. ➢ Constant monitoring for proactive defence against cyber threats.
Offensive Capabilities
➢ Deploying malware and ransomware to disable enemy systems. ➢ Phishing attacks to disrupt enemy communications.
Defensive Strategies
➢ Advanced firewalls and intrusion detection systems to protect sensitive data. ➢ Secure communication protocols for military operations.
India's Efforts
➢ Indigenous cybersecurity frameworks by DRDO and CERT-In. ➢ Development of Netra, an internet surveillance system.

10. **Additive Manufacturing**. Additive manufacturing, commonly known as 3D printing, has emerged as a valuable technology in military logistics and production. It allows for on-demand production of essential parts, reducing logistical demands and enabling rapid prototyping of new equipment. This technology also opens possibilities for creating advanced materials, such as self-healing Armour and temperature-controlled clothing for soldiers.



3D PRINTED BUNKER IN LADDAKH



3D PRINTED ATC TOWER

SPECIFICATIONS	
TECHNOLOGY	3D printing (metal and polymer-based)
APPLICATIONS	Ammunition, spare parts, equipment, clothing
MATERIALS	Metal alloys, thermoplastics, high-strength polymers
CAPABILITIES	Customization, rapid prototyping, localized production
ADVANTAGES	Cost-effectiveness, reduced logistical strain, enhanced adaptability
Features and Capabilities	
➢ Reduces the need for large stockpiles by enabling on-demand production. ➢ Capable of creating specialized ammunition and military gear. ➢ Supports innovative materials like self-healing armour and temperature-regulated clothing. ➢ Allows for localized manufacturing, decreasing dependence on supply chains. ➢ Facilitates repair and maintenance by producing replacement parts in the field.	
Applications	
➢ Localized production of spare parts and equipment. ➢ Creation of self-healing armour and lightweight materials.	
Advantages	
➢ Improves decision-making with a unified command network. ➢ Reduces response time during critical missions.	
India's Use	
➢ Reduces production costs and time. ➢ Enables on-demand manufacturing in remote areas.	

PART II: EMERGING TECHNOLOGIES AND FUTURE INDUCTIONS

11. The evolution of modern warfare necessitates continuous innovation in weapon systems to address emerging threats and enhance battlefield effectiveness. Conceptualized weapon systems represent the cutting edge of defence technology, encompassing ideas and designs that are either under development in prototype stages or undergoing trials.

12. These systems are conceptualized to meet specific operational requirements, such as improving precision, range and survivability or integrating advanced technologies like

Artificial Intelligence, robotics and directed energy. They often focus on addressing future warfare scenarios, including network-centric operations, asymmetric threats and high-intensity conflicts.

13. Through rigorous testing and iterative improvements, these prototypes are refined to deliver superior performance and reliability. Once successful, these systems pave the way for the next generation of weapons shaping the future of military capabilities and strategy.

14. **FRCV (Future-Ready Combat Vehicle)**. The Future-Ready Combat Vehicle (FRCV) is an initiative by the Indian Army to replace the ageing T-72 tank fleet which has been the backbone of India's armoured capabilities for decades. With the modern battlefield becoming increasingly complex and technologically advanced, there is a need for a tank that can match global standards in terms of firepower, mobility, and protection. The FRCV program is focused on developing a next-generation main battle tank (MBT) with improved capabilities to face contemporary threats.

SPECIFICATIONS	
ARMOUR	Composite and modular Armour with additional active protection systems.
FIREPOWER	Advanced main gun with enhanced penetration capabilities and support for guided munitions.
MOBILITY	High-speed, all-terrain capability with a powerful engine for quick deployment.
ELECTRONICS	Equipped with digital fire control systems, AI-driven threat detection, and night vision.
CREW	3-person crew with enhanced ergonomic design for operational efficiency.
Features and Capabilities	
➢ Enhanced Protection: Multi-layered Armour with modular components allows for adaptability against varied threats. ➢ Improved Firepower: Capable of firing high-velocity shells and guided missiles, increasing its range and accuracy. ➢ Advanced Mobility: Equipped with a powerful engine and enhanced suspension for rough terrains and fast mobilization. ➢ Automation: Integration of AI systems for situational awareness and targeting, reducing human error. ➢ Examples in Use: The FRCV is intended to surpass current T-90 tanks in performance, positioning India on par with international tank platforms like the U.S. Abraham and the German Leopard 2.	

15. **ATAGS (Advanced Towed Artillery Gun System)**. The Advanced Towed Artillery Gun System (ATAGS) is an indigenous project to upgrade India's artillery capabilities by replacing old howitzers. Developed by the Defence Research and Development Organisation (DRDO) in collaboration with private firms, ATAGS is designed to provide long-range, accurate, and effective artillery support to ground forces.



SPECIFICATIONS	
CALIBER	155 mm, 52-caliber barrel
RANGE	Capable of firing up to 48 kilometers depending on ammunition type
MOBILITY	Designed for rapid deployment with towability across varied terrains
AUTOMATION	Semi-automated ammunition loading system for faster firing rates
WEIGHT	Approximately 18 tons, suitable for transportation by military trucks
EXTENDED RANGE	ATAGS has one of the longest ranges in its class, allowing for effective coverage of extensive battlefield
HIGH RATE OF FIRE	The semi-automated loading system enables quick firing, crucial for sustained artillery bombardment
ACCURACY AND RELIABILITY	Advanced fire control systems ensure high precision and durability.
INDIGENOUS DESIGN	Boosts India's self-reliance in defence manufacturing, with all components developed domestically.
EXAMPLES IN USE	Once fully operational, ATAGS will replace the existing Bofors howitzers, significantly enhancing India's artillery reach and firepower.

16. **CADET (Carrier Air Defence Tracked System)**. CADET (Carrier Air Defence Tracked System) is a self-propelled air defence system intended to protect armoured columns from aerial threats. Integrated with India's 'Akashteer' Air Defence Control and Reporting System, CADET offers an effective mobile shield against enemy aircraft, drones, and missiles. This system is essential for the Indian Army's future mechanized warfare, providing close air defence support for moving troops.

SPECIFICATIONS	
WEAPONRY	Equipped with short-range surface-to-air missiles (SAMs)
MOBILITY	Tracked system designed to keep pace with armoured divisions
INTEGRATION	Linked with Akashteer AD system for real-time air threat information
RADAR	On-board radar with a 360-degree tracking capability
CREW	Operated by a small crew with automated firing and targeting systems.

Features and Capabilities

- Mobile Air Defence: Provides defence for advancing forces against low-flying aircraft and UAVs.
- Real-Time Targeting: Integration with Akashteer allows for synchronized defence with other air defence assets.
- Rapid Deployment: Tracked chassis enables movement in difficult terrains alongside tank regiments.
- Enhanced Survivability: Equipped with electronic countermeasures to resist jamming and other interference.
- Examples in Use: The CADET will offer India a capability similar to the Russian TOR system, providing mobile SAM support to frontline units.

17. **Sudarshan CIWS (Close-in Weapon System)**. The Sudarshan Close-In Weapon System (CIWS) is designed as a last line of defence against incoming aerial threats such as missiles, aircraft, and drones. This system plays a crucial role in protecting high-value assets and installations by intercepting and neutralizing threats in close proximity.

SPECIFICATIONS	
WEAPONRY	Multi-barrel guns with high rate of fire.
TRACKING	Advanced radar and electro-optical tracking for precise targeting.
RANGE	Effective within a few kilometers to protect critical areas.
RESPONSE TIME	Capable of rapid response to multiple simultaneous threats.
OPERATION	Automated engagement with manual override options.

Features and Capabilities

- Rapid-Fire Defence: Designed to neutralize close-range threats with a very high rate of fire.
- 360-Degree Protection: Capable of tracking and engaging threats from any direction.
- Advanced Targeting: Combines radar and optical tracking for accuracy.
- Examples in Use: Like the U.S. Phalanx CIWS, the Sudarshan CIWS will enhance India's point-defence capabilities, especially around critical installations.

18. **HAL's Prachand/Rudra Attack Helicopters**. The Hindustan Aeronautics Limited (HAL) "*Prachand"* and "*Rudra"* helicopters are indigenous attack helicopters developed to meet the Indian military's need for agile, versatile, and lethal air support. They are designed to perform both reconnaissance and combat missions, equipped with weaponry to engage ground and aerial targets.



SPECIFICATIONS	
ARMAMENT	Air-to-ground missiles, rockets, and machine guns.
AVIONICS	Advanced navigation, targeting, and surveillance systems.
RANGE	Capable of extended operational range with in-flight refueling.
CREW	Two pilots with a dual-control cockpit.
MOBILITY	Excellent manoeuvrability for high-altitude operations.

Features and Capabilities

- Multi-role Capabilities: Suitable for both attack and reconnaissance missions.
- High-Altitude Performance: Operable in mountainous terrains, particularly in border regions.
- Heavy Armament: Equipped with a range of missiles and rockets for diverse combat roles.
- Examples in Use: Deployed along India's northern and western borders, enhancing India's air assault capabilities against both ground targets and light armoured vehicles.

19. **MPATGM (Man-Portable Anti-Tank Guided Missile)**. The Man-Portable Anti-Tank Guided Missile (MPATGM) is India's next-generation, lightweight, and highly effective anti-tank weapon system. Designed to replace the aging '*Konkur'* and '*Milan'* systems, it empowers Infantry units with the capability to neutralize armoured threats in close-quarters and high-mobility scenarios.



SPECIFICATIONS	
RANGE	Effective up to 2.5 kilometers
GUIDANCE	Fire-and-forget with infrared homing capability
WEIGHT	Portable with minimal setup requirements
WARHEAD	Tandem warhead effective against reactive armour
DEPLOYMENT	Operable by a single soldier with minimal preparation time
Features and Capabilities	
➢ Lightweight and Portable: Suitable for Infantry units requiring high mobility. ➢ Advanced Guidance System: Infrared homing for 'fire-and-forget' capability. ➢ Enhanced Armour Penetration: Tandem warhead penetrates even reactive armour. ➢ Examples in Use: Designed to give Infantry a comparable system to the American Javelin, enhancing anti-tank capabilities in high-intensity conflict scenarios.	

20. **Drones and their use in Modern**. Warfare Drones have become indispensable in modern warfare, offering capabilities that enhance reconnaissance, surveillance, and strike precision. These unmanned aerial vehicles (UAVs) can operate in high-risk areas, gather real-time intelligence and in some cases execute targeted strikes. India is increasingly integrating drones into its defence strategy to improve situational awareness and reduce risk of loss of life of fighting troop.

SPECIFICATIONS	
TYPES	Reconnaissance drones, combat drones and loitering munitions.
RANGE	Varied, depending on mission type (from short-range to beyond line of sight).
PAYLOAD	Cameras, sensors, missiles, explosives.
CONTROL	Remote or autonomous navigation.
FLIGHT DURATION	Depends on model and mission requirements.

Features and Capabilities

- Real-Time Intelligence Gathering: Provides continuous monitoring of target areas
- Surveillance: Capable of high-altitude, long-duration flights for area mapping
- Precision Strike: Equipped with missiles or loitering munitions for targeted attacks
- Reduced Troop Risk: Operates in high-risk zones, minimizing casualties
- Adaptability: Suitable for reconnaissance, electronic warfare, and direct combat.

CONCLUSION

21. As the cadets have seen, technology is the future of Defence Forces in India. India has taken the foot forward and a plethora of weapons and equipment is being inducted. Be it submarines, Long and Short Range Missiles, Fifth Generation Fighter Jets or Rifles for the Infantry, the next decade will see a huge defence technology. The Defence sector manufacturing will increase manifolds with huge investments coming from the established industries in the country.

SUMMARY

Part I : Advanced Military Technologies:-

- **Directed Energy Weapons (DEWs)**.
  - Use focused energy beams like lasers or microwaves to neutralize targets with high precision.
  - Key Example: DRDO's DURGA-II, designed to counter drones and missiles.
- **Precision Guided Munitions (PGMs)**.
  - Enable targeted strikes with minimal collateral damage using advanced guidance systems (GPS, laser).
  - Key Examples: Sudarshan laser-guided bombs, HELINA missiles.
- **Loitering Munitions**.
  - Combine drone and missile technologies for real-time surveillance and precision strikes.
  - Key Example: Harop drones for destroying radar systems.
- **AI-Based Situational Awareness Systems**.
  - Utilize AI for reconnaissance, surveillance, and decision-making.
  - Key Example: DRDO's CIDSS for enhanced battlefield intelligence.
- **Military Robotics and Autonomous Systems**.
  - Robots for reconnaissance, logistics, and bomb disposal, minimizing human risk.
  - Key Example: Daksh robot for explosive ordnance disposal.
- **Internet of Military Things (IoMT)**.
  - Network of interconnected military assets for real-time battlefield awareness.
  - Key Example: Tactical Communication Systems (TCS) in India.
- **Cyber Warfare**.
  - Offensive and defensive strategies to secure digital infrastructure and disrupt enemy networks.
  - Key Example: DRDO's Netra for internet surveillance.
- **Additive Manufacturing (3D Printing)**
  - Enables localized production of military equipment, reducing logistical burdens.
  - Key Example: DRDO's hubs for rapid prototyping of military components.

Part II: Emerging Technologies and Future Inductions:-

- **FRCV (Future-Ready Combat Vehicle)**. Next-generation tank to replace T-72 with advanced mobility and firepower.
- **ATAGS (Advanced Towed Artillery Gun System)**. High-range artillery system developed indigenously.
- **CADET (Carrier Air Defence Tracked System)**. Mobile air defence system integrated with Akashteer for frontline units.
- **Sudarshan CIWS**. Close-in weapon system for neutralizing aerial threats.
- **HAL's Prachand/Rudra Helicopters**. Multi-role attack helicopters for high-altitude operations.
- **MPATGM (Man-Portable Anti-Tank Guided Missile)**. Lightweight anti-tank system with fire-and-forget capability.
- **Immersive Technologies and Augmented Realities**. Wearable devices providing real-time battlefield data and 3D terrain maps.
- **Drones in Modern Warfare**. Enhance reconnaissance, surveillance, and strike capabilities.

ASSESSMENT EXERCISE

Multiple Choice Questions (MCQs)

Q1. What is the key feature of Directed Energy Weapons (DEWs)?

(a) Use of explosives
(b) Use of focused energy beams
(c) Autonomous operation
(d) GPS-guided targeting

Q2. Which system is developed by DRDO for situational awareness?

(a) Daksh
(b) CIDSS
(c) FRCV
(d) ATAGS

Q3. What is the primary use of loitering munitions?

(a) Reconnaissance only
(b) Real-time surveillance and precision strikes
(c) Logistics support
(d) Air Defence

Q4. Which missile is a man-portable anti-tank guided weapon?

(a) BrahMos
(b) MPATGM
(c) HELINA
(d) Harop

Q5. What does IoMT stand for?

(a) Internet of Military Technologies
(b) Internet of Military Things
(c) Integrated Military Technology
(d) Intelligent Military Tracking

Q6. Which Indian-developed system is designed to replace aging T-72 tanks?

(a) ATAGS (b) FRCV
(c) Prachand (d) Rudra

Q7. What is the primary function of the Advanced Towed Artillery Gun System (ATAGS)?

(a) Anti-aircraft defence
(b) High-range artillery strikes
(c) Cybersecurity
(d) Precision bombing

Q8. Which type of ammunition combines drone and missile technologies?

(a) Precision Guided Munitions
(b) Loitering Munitions
(c) Ballistic Missiles
(d) Anti-Tank Missiles

Q9. Which of the following is NOT a feature of Internet of Military Things (IoMT)?

(a) Interconnected battlefield assets
(b) Real-time data sharing
(c) Enhances battlefield situational awareness
(d) Missile launching capability

Q10. Which drone is specifically used for reconnaissance by the Indian Army?

(a) Harop
(b) Heron
(c) Switchblade
(d) Daksh

Q11. What does the Sudarshan CIWS replace?

(a) Ballistic missiles
(b) Traditional anti-aircraft guns
(c) Armoured vehicles
(d) Artillery systems

Q12. Which robot is used for explosive ordnance disposal in India?

(a) Rustom
(b) Daksh
(c) CIDSS
(d) Prachand

Q13. What makes additive manufacturing advantageous in military applications?

(a) High-speed mobility
(b) Lightweight armour production
(c) Cost-effective and on-demand production
(d) Real-time data sharing

Q14. Which missile system is launched from helicopters?

(a) BrahMos
(b) HELINA
(c) MPATGM
(d) ATAGS